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APPLICATION EXAMPLE

Redundant IO modules connected to SIMATIC S7-1500

LRedIo Library V2.0

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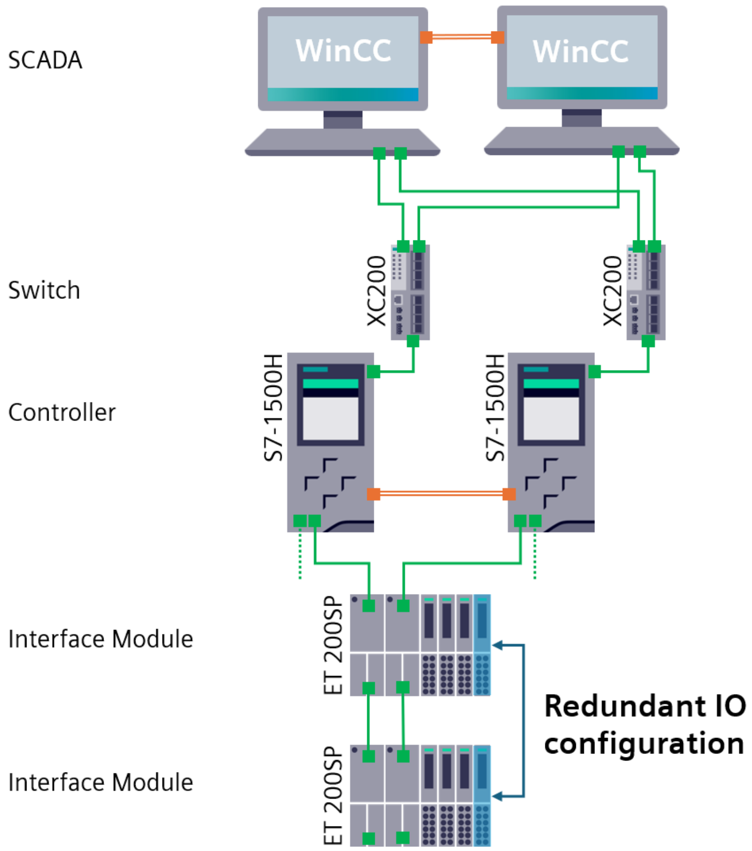
1. Introduction

1.1. Use case scenario

Even the use of industry-proven and reliable components, failures in the production line can happen, and potentially having a critical impact on the production. Therefore, manufacturers need to assess the system availability to identify single points of failures and find solutions to reduce risk of downtime. By introducing redundancies, the availability of the automation system can significantly be increased.

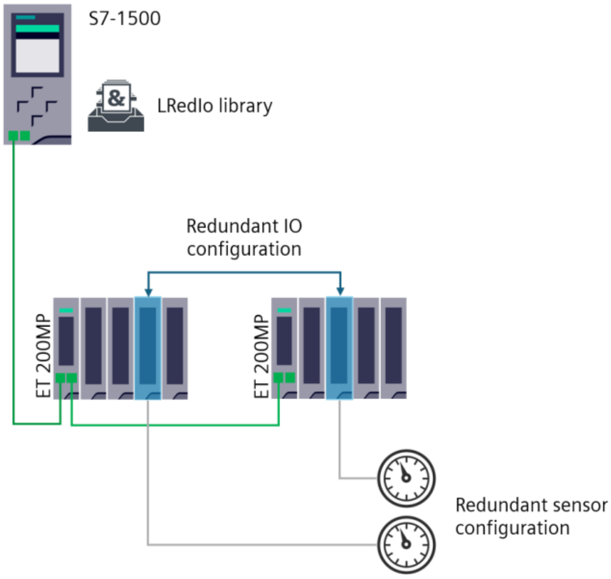
Siemens offers tailored redundancy solutions for each part of the automation system:
SCADA, switches, controller, PROFINET interface module, IO-modules and power supply.



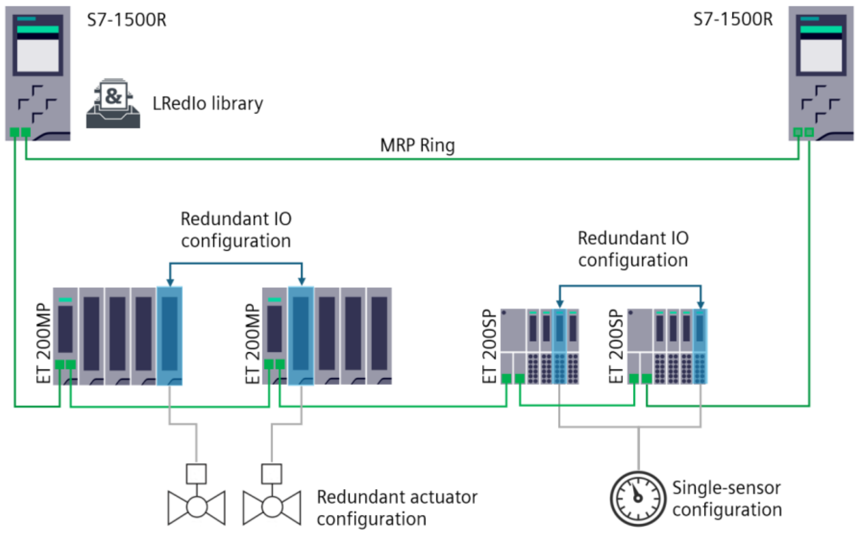
If the availability of sensor and/or actuator loops needs to be enhanced, this application offers effective solutions.
By incorporating redundancy of the IO-modules connected to single or redundant sensors/actuators, the availability of these loops will be significantly improved. Thanks to the monitoring performed by the LRedIo function blocks, even in the event of an IO-module or channel failure, the automation system will continue to operate, ensuring that the customer process remains unaffected.

1.2. Configuration examples

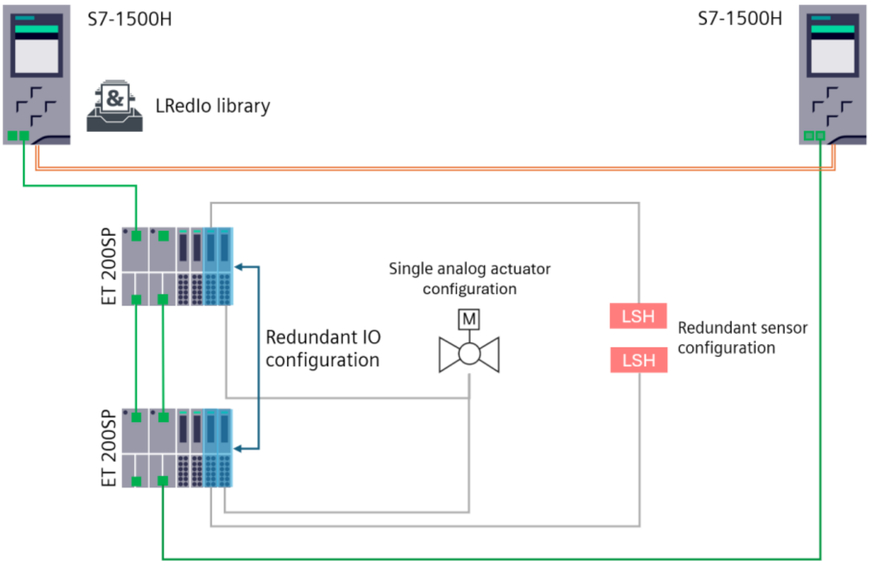
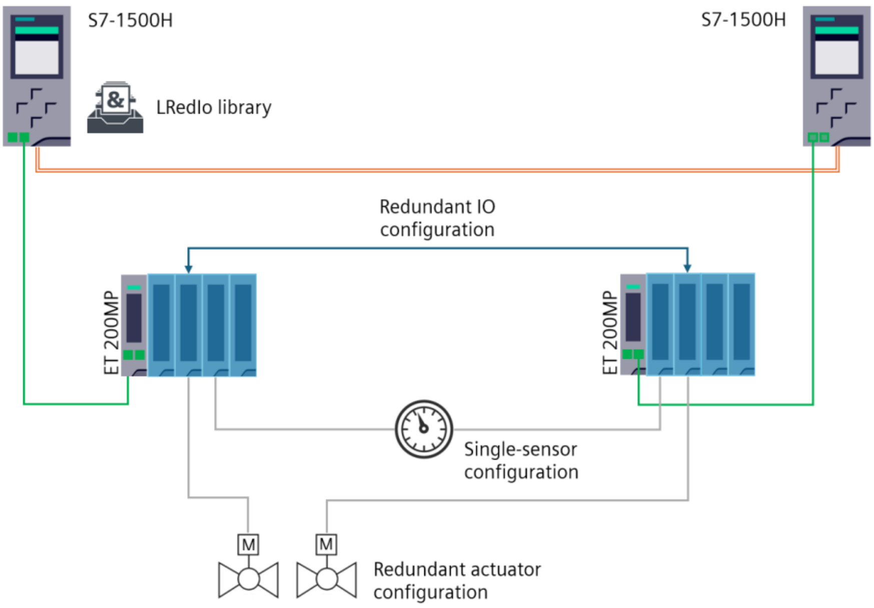
1.2.1. With single S7-1500 controller

Configuration	Description
	Single S7-1500 controller Some IO-modules have a redundant setup. Highest Availability of the **sensor loop** is required: two sensors / two AI modules (one pair).

1.2.2. With S7-1500R redundant system

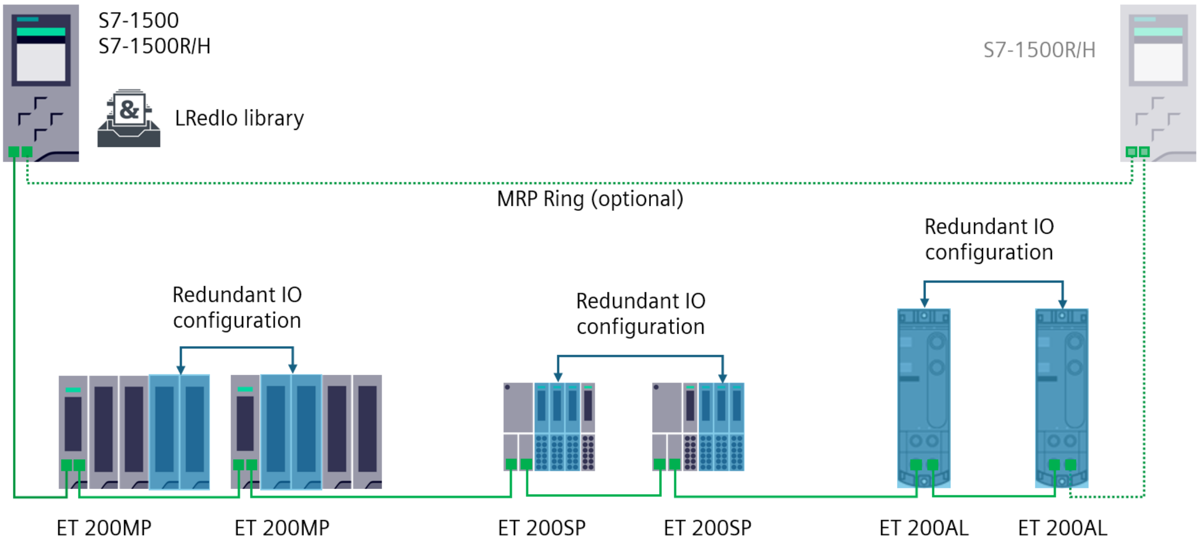
Configuration	Description
	S7-1500R redundant system (MRP Ring topology). S2 Devices (single PROFINET interface module), some IO-modules have a redundant setup. Highest Availability of the **reaction loop** is required: two actuators / two DQ modules (one pair). Highest Availability of the **sensor acquisition** is required: single sensor / two AI modules (one pair).

1.2.3. With S7-1500H redundant system

Configuration	Description
	S7-1500H redundant system (Line topology). R1 devices (redundant PROFINET interface module), some IO-modules have a redundant setup. Highest Availability of the **sensor loop** is required: two LSH sensors (Level Switch High) / two DI modules (one pair). Highest Availability of the **reaction** is required: single analog actuator / two AQ modules (one pair).
	S7-1500H redundant system (Line topology). Redundant S1 devices (single PROFINET interface module). With such configuration, a full module redundancy is mandatory. High Highest Availability of the **sensor acquisition** is required: single analog sensor / two AI modules (one pair). Highest Availability of the **reaction loop** is required: two analog actuators / two AQ modules (one pair).

1.3. Technical Solution

The principle involves using a pair of IO modules (with same MLFB and firmware). These modules can be located within the same station or across two different stations. A channel of each IO-module is wired to a single or redundant sensors/actuators. Using a function block of the LRedIo library, the consistency and quality of both signals are continuously monitored, providing valuable diagnostic information for the redundant module.



The LRedIo library contains standard Function Blocks (FB) for the implementation of redundant IO-module applications using CPUs of the SIMATIC S7-1500. This library is supported by all SIMATIC S7-1500 CPUs, are included also the redundant controllers S7-1500R/H.

Function block	Description
LRedIo_RedDi	Redundancy function for two digital inputs
LRedIo_RedAi	Redundancy function for two analog inputs
LRedIo_RedDq	Redundancy function for two digital outputs
LRedIo_RedAq	Redundancy function for two analog outputs

These Function Blocks are provided as a library and can be used freely. The completed functions are fully customizable, allowing for unrestricted use.

1.4. Hardware and software requirements

Software Requirements

The function blocks of the LRedIo library have been tested with TIA Portal V19. However, there is no restriction on using them with other TIA Portal versions. For the TIA Portal versions higher than V19, the LRedIo library can be updated accordingly. For the version lower than V19, the source files (.scl) are available and provided for that purpose.

Hardware – Controller requirements

The function blocks of the LRedIo library have been tested with a S7-1500 firmware version 2.9. But there is also no restriction on using them with a different firmware version.

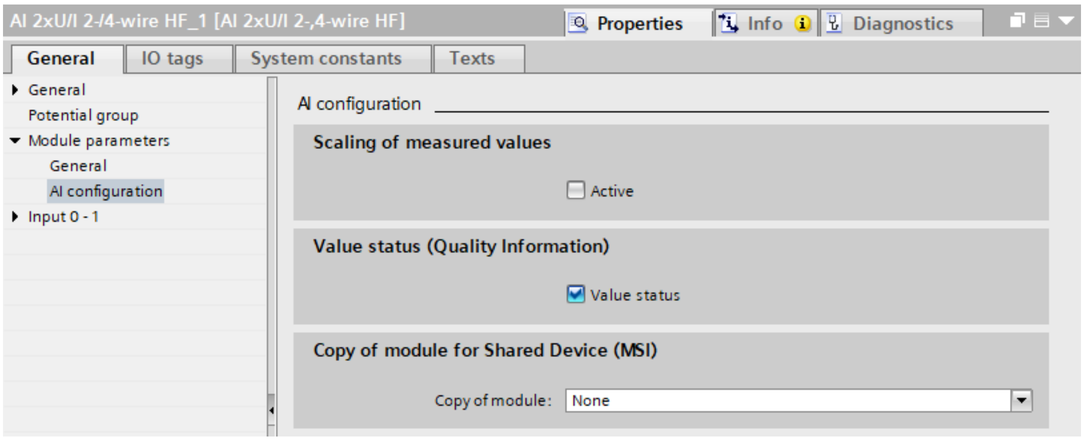
Hardware – IO-module requirements

The modules must provide a Value Status (Quality information).
For the pair of IO modules, the MLFB and firmware version should be the same.

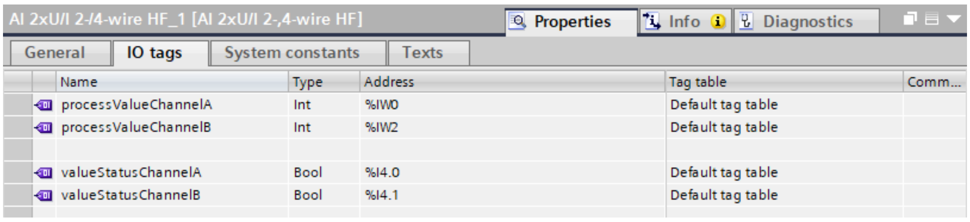
1.5. Value Status

To evaluate the quality of the IO-channel used, the IO-modules must provide quality information, so-called Value Status. Each Value Status bit is assigned to a channel and provides information on the quality of the channel (FALSE = incorrect or bad quality, e.g., wire break).

The Value Status must be enabled in the module configuration as the following example.

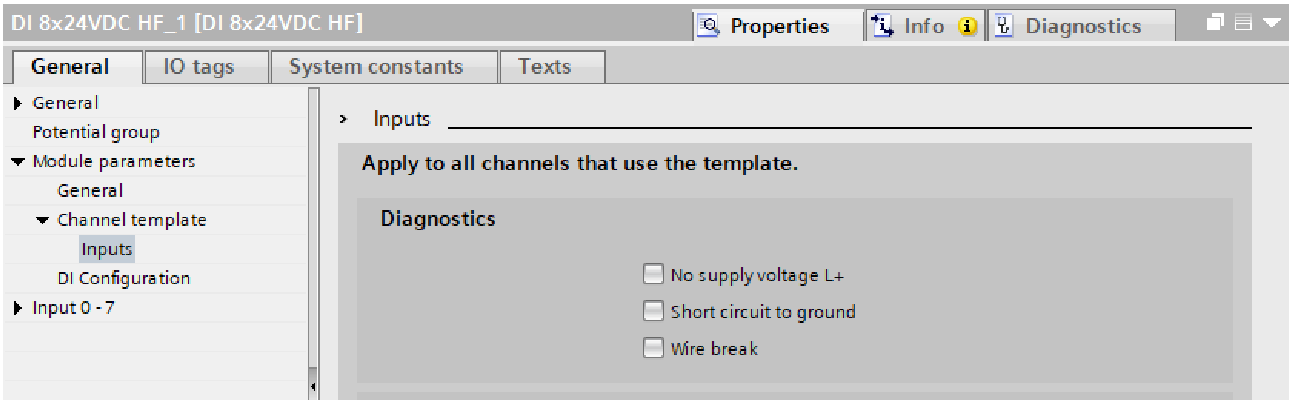


As consequences, for each process value, a Value Status bit is available and will be evaluated by the corresponding LRedIo function block.



1.6. Diagnostic and Value Status

If the Value Status is enabled, the module always checks all errors even if the respective diagnostic message is not enabled. As example, diagnostic messages available by a DI module:



The channel diagnostic configuration is a selection of diagnostic message required by the application but does not affect the Value Status behavior. Regardless of whether diagnostic messages are enabled or disabled, the Value Status will react in the event of an error.

Example:

A wire break error occurs and the diagnostic message "Wire break" is disabled, then no diagnostic message will be reported, but the Value Status bit will switch to FALSE (0).

NOTE	In case of error, the Value Status will always react regardless of the channel diagnostic configuration.

2. IO-modules

2.1. ET 200SP/MP

It must be mentioned that with ET 200SP as well as ET 200MP several categories of IO-modules are available: Standard (**ST**), High Feature (**HF**) and High speed (**HS**).

Depending on this category, the IO-modules provide diagnostics per load group / module or for each channel.
See [What are the differences between the Basic (BA), Standard (ST), High Feature (HF) and High Speed (HS) modules of the ET 200SP and ET 200MP?]

Diagnostic per load group / module:

If one channel fails, then all the channels of the same load group or the complete module will deliver a Value Status of FALSE. Other channels are affected by a single channel failure. The unused channels must be deactivated, otherwise they could trigger an error that affects all other channels of that module.

Diagnostic for each channel:

If one channel fails, then only the affected channel will deliver a Value Status of FALSE. No impact on the other channels. This means that unused channels do not necessarily have to be deactivated.

	DI/DQ	AI/AQ	Diagnostic level
ET 200SP – Standard	✓	✓	Diagnostic per load group or module
ET 200SP – High Feature / High Speed	✓	✓	Diagnostic for each channel
ET 200MP – Standard Digital modules	✓		Diagnostic per load group or module
ET 200MP – Standard Analog modules		✓	Diagnostic for each channel
ET 200MP – High Feature / High Speed	✓	✓	Diagnostic for each channel

NOTE For a highest availability solution, the diagnostic per channel is highly recommended.

2.2. ET 200AL

With ET 200AL, the diagnostic level depends on the module types.

All digital modules (DI/DQ) offer a diagnostic per load group / module:

If one channel fails, then all the channels of the same load group or the complete module will deliver a Value Status of FALSE. Other channels are affected by a single channel failure. The unused channels must be deactivated, otherwise they could trigger an error that affects all other channels of that module.

All analog modules (AI/AQ) offer a diagnostic for each channel:

If one channel fails, then only the affected channel will deliver a Value Status of FALSE. No impact on the other channels. This means that unused channels do not necessarily have to be deactivated.

	DI/DQ	AI/AQ	Diagnostic level
ET 200AL –Digital modules	✓		Diagnostic per load group or module.
ET 200AL –Analog modules		✓	Diagnostic for each channel.

2.3. ET 200eco PN M12-L

With ET 200eco PN M12-L, all IO-modules offer a diagnostic for each channel.

2.4. ET 200clean

With ET 200clean, all IO-modules offer a diagnostic for each channel.

2.5. ET 200SP HA

With ET 200SP HA, all IO-modules offer a diagnostic for each channel.

NOTE The SIMATIC ET 200SP HA terminal blocks with 2 slots for IO redundancy are not supported in TIA Portal (e.g., 6DL1193-6TC00-0DR0). Therefore, they cannot be used.

2.6. Not-suitable IO-modules

The following table shows the modules which are NOT suitable for the usage of LRedIo library.
The Value Status quality information is NOT available.

Not suitable	Reason
ET 200MP – Basic version	With BA modules, no Value Status
ET 200SP – Basic version	With BA modules, no Value Status
ET 200PRO	No Value Status
ET 200iSP	No Value Status
ET 200eco PN M12	No Value Status

2.7. Single sensor/actuator - IP20 vs. IP6x

For setups involving a single sensor and/or actuator, additional components like Zener diodes are required.

In IP20 installations, cabinets are already present, allowing these components to be conveniently installed near the IP20 IO-stations, such as a station ET 200SP/MP.

However, in IP6x installations, which are by design cabinet-free, finding a suitable location for these components can be challenging.

Consequently, even if it is technically feasible to use such single senor/actuator configurations in an IP6x environment, the focus in this document will be on IP20 IO-stations.

3. Redundant Digital Input

3.1. Use cases

Redundant digital input module pair with:

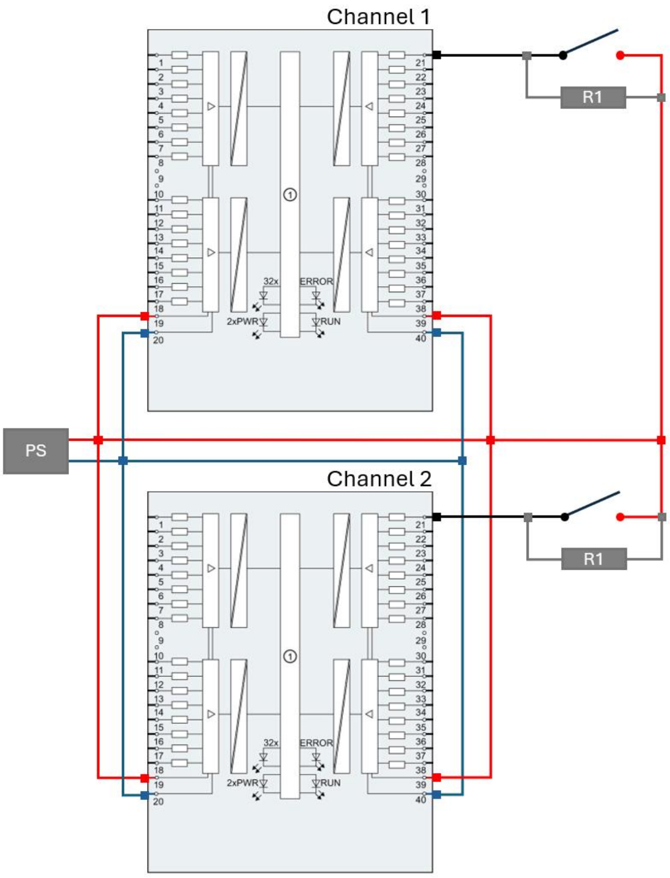
- a configuration two sensors (wire break detection with additional resistor R1 required),
- a configuration single sensor (wire break detection with additional resistor R1 required).

3.2. Wiring configurations

3.2.1. Configuration two sensors

Wiring example based on ET 200MP DI 32x24VDC HF (6ES7521-1BL00-0AB0):

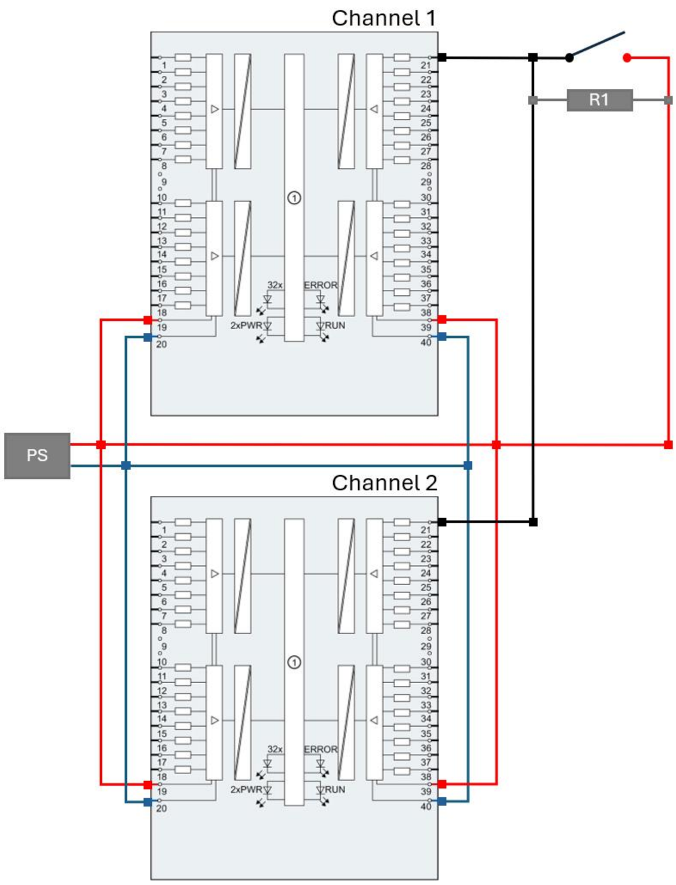
(Wire break detection with additional resistor R1 required)



3.2.2. Configuration single sensor

Wiring example based on ET 200MP DI 32x24VDC HF (6ES7521-1BL00-0AB0):

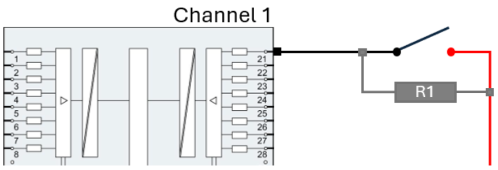
(Wire break detection with additional resistor R1 required)



3.3. Wire break

If the Digital Input module used supports the wire break detection, then an additional resistor R1 is required.
Even the diagnostic message "Wire break" is not enabled, the Value Status will react in case of Wire break detection.

Connect a resistor (R1) in parallel of the sensor:



According to the manuals, suitable resistance (R1) for wire break detection:

- ET 200SP and ET 200MP: resistor of 25kΩ to 45kΩ.
- ET 200SP HA: resistor of 15kΩ to 18kΩ.
- Even the integration of resistor into IP6x installation may be problematic, there is also solution with for example ET 200eco PN M12-L and a resistor of 25kΩ to 45kΩ.

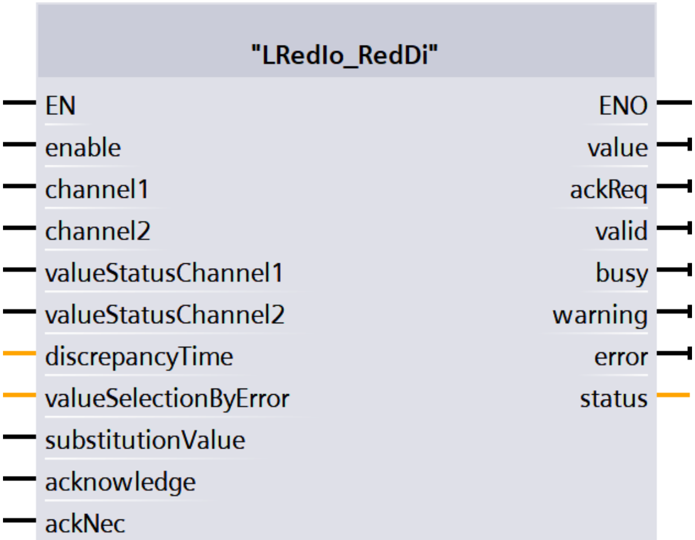
NOTE To get more diagnostic information, enable the wire break diagnostic message in the hardware configuration of the considered channel.

3.4. LRedIo_RedDi

Short description

The function block "LRedIo_RedDi" is used for the evaluation of redundant digital input channels within the user program. According to the process value and the Value Status of each digital input channel, the block provides a process value which is used in the user program. The consistency as well as the quality of both signals are cyclically monitored and therefore deliver valuable diagnostic information of the redundant digital input module pair.

Block interface



Input parameter

Name	Data type	Description
enable	Bool	TRUE: enable functionality of FB
channel1	Bool	Channel 1
channel2	Bool	Channel 2
valueStatusChannel1	Bool	Value Status of the channel 1
valueStatusChannel2	Bool	Value Status of the channel 2
discrepancyTime	Time	Max. discrepancy time between the inputs (min. 20ms - max. 500ms)
valueSelectionByError	USInt	By error selection of the value used 0 = Passivated (FALSE) 1 = Substitute value 2 = Last valid value
substitutionValue	Bool	Substitution value
acknowledge	Bool	Acknowledgement and reintegration after startup or invalid Value Status
ackNec	Bool	TRUE: Acknowledgement necessary

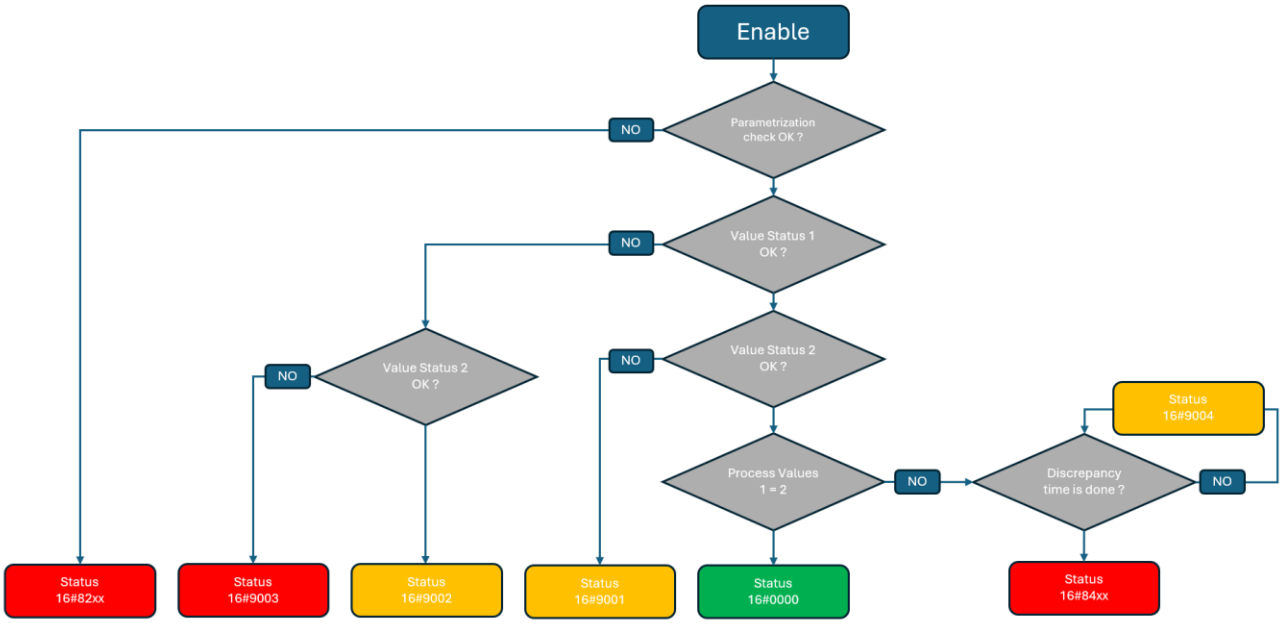
Output parameter

Name	Data type	Description
value	Bool	Evaluated input signal
ackReq	Bool	TRUE: acknowledgement request
valid	Bool	TRUE: valid set of the value parameter
busy	Bool	TRUE: FB is running
warning	Bool	TRUE: a channel is invalid
error	Bool	TRUE: an error occurred during the execution of the FB
status	Word	16#0000 - 16#7FFF: Status of the FB, 16#8000 - 16#FFFF: Error identification

Status information

Name	Description
16#0000	Both channels are valid
16#7000	No job being currently processed
16#7001	First call after incoming new job (rising edge 'enable')
16#7002	Subsequent call during active processing without further details
16#7003	Block waits for initial acknowledge
16#8205	Parametrization error at substitution value
16#8206	Parametrization error at discrepancy time
16#8400	Discrepancy error with channel passivation
16#8401	Discrepancy error with substitution value
16#8402	Discrepancy error with last valid value
16#8600	Error due to an undefined state in state machine
16#9001	Invalid value on Value Status 2
16#9002	Invalid value on Value Status 1
16#9003	Invalid value on both channels
16#9004	Discrepancy between channels but disc time still running

Functional description



Internal block description

State	Comments
Parametrization error	The function block checks if the parametrization is correct. "**discrepancyTime**" > 20ms and < 500ms "**valueSelectionByError**" between 0 and 2 Otherwise a parametrization error is triggered.
Normal operation	When there is no channel error ("Value Status" of both channels have the value "TRUE") and there is no discrepancy error (same value for "**channel1**" and "**channel2**", either "FALSE" or "TRUE"), then the output "**value**" will be controlled using the value of "channel1".
Value Status	When only one of the channels reports an error ("**valueStatusChannel1**" = "FALSE" or "**valueStatusChannel2**" = "FALSE"), then the output "**value**" will be controlled using the value of the errorfree channel. When both channels report an error ("**valueStatusChannel1**" = "FALSE" and "**valueStatusChannel2**" = "FALSE"), then the output "**value**" depends on the parameter "**valueSelectionByError**" defined by the user: "0" = passivated (FALSE), "1" = Substitute value or "2" = the last valid value will be used.
Behavior by discrepancy	When both channels do not report any error but there is a discrepancy between the "**channel 1"** and the "**channel 2**" longer than the discrepancy time, then an error is triggered and the output "**value**" will be set according to the user configuration "**valueSelectionByError**": "0" = passivated (FALSE), "1" = Substitute value or the last valid value is used.
Acknowledgement	As soon as both channels do not report any error anymore and have the same value (no discrepancy), then the output "**ackReq**" is set to "TRUE". A rising edge on the input **"acknowledge"** will acknowledge the error. An acknowledgment is only needed after an error when the parameter "**ackNec**" is set to "TRUE", otherwise no user action is required.

4. Redundant Digital Output

4.1. Use cases

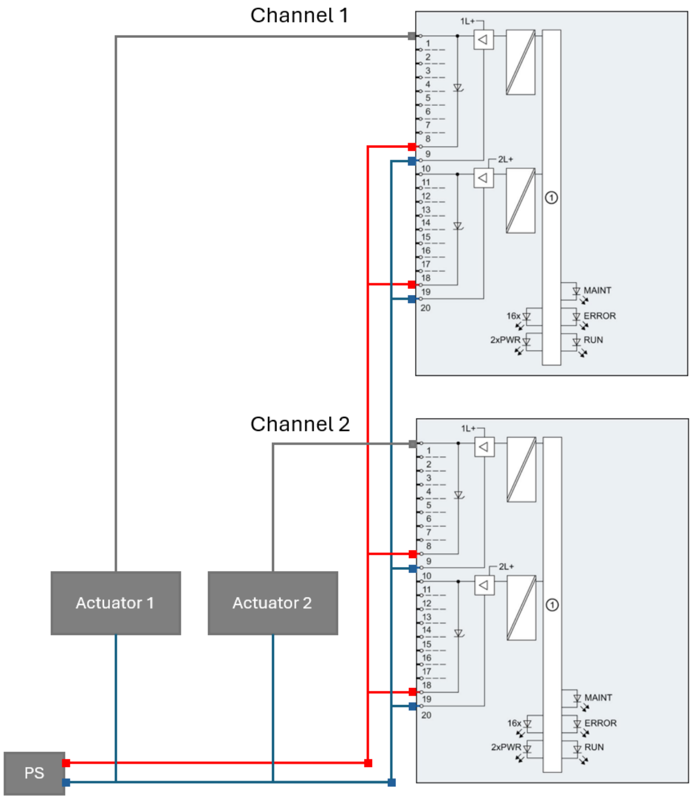
Redundant digital output module pair with:

- a configuration two actuators
- a configuration one actuator (additional diodes are required)

4.2. Wiring configurations

4.2.1. Configuration with two actuators

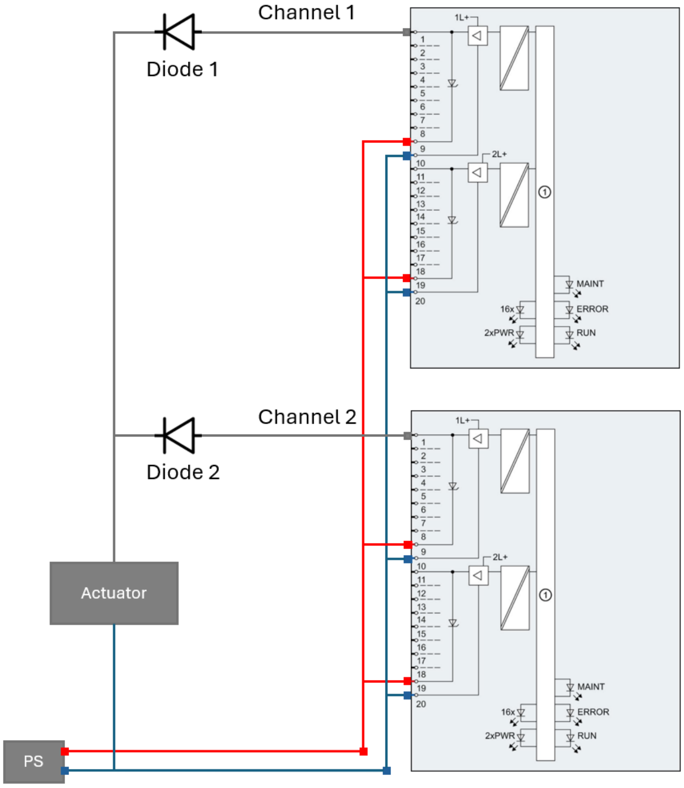
Wiring example based on ET 200MP DQ 16x24VDC/0.5A HF (6ES7522-1BH01-0AB0)



4.2.2. Configuration with single actuator

Wiring example based on ET 200MP DQ 16x24VDC/0.5A HF (6ES7522-1BH01-0AB0)

NOTE To allow a reintegration after error, the diagnostic message wire-break must be disabled for both channels.
Even the diagnostic message wire-break is disabled, this error is still detected, and the Value Status of the channels will react in consequence, but no entry will be generated in the diagnostic buffer and the channel will not report this error by flashing the led in red.



4.3. Single actuator and diode

The table below shows the Digital Output module families which in redundant mode need additional diodes:

ET 200 Family (only IP20)	Additional diode	Comments
ET 200SP	Yes	For example, through terminal blocks
ET 200SP HA	No	Included by design into the DQ module
ET 200MP	Yes	For example, through terminal blocks

For ET 200SP or ET 200MP, suitable diodes are with:

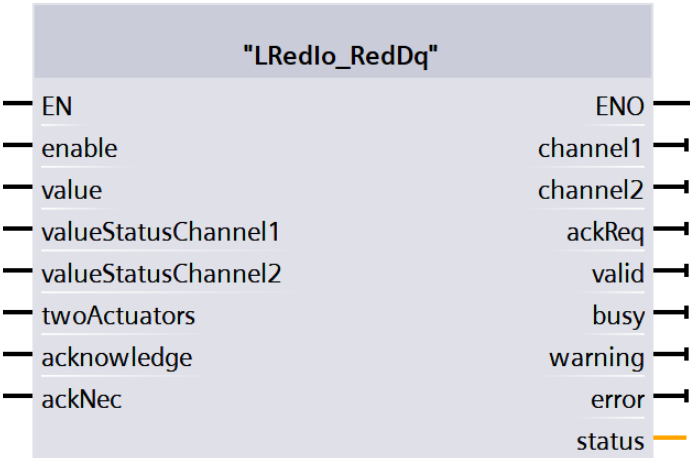
- **Voltage Rating:** maximum repetitive peak reverse voltage (V_{RRM}) of > 200V,
- **Current Handling:** forward continuous current (I_F) >= 1A.
- Example: **Type 1N4007**

4.4. LRedIo_RedDq

Short description

The function block “LRedIo_RedDq” is used for the evaluation of redundant digital output channels within the user program. The block forwards the process value calculated by the user program to both digital output channels. The quality of both signals is cyclically monitored and therefore deliver valuable diagnostic information of the redundant digital output module pair.

Block interface



Input parameter

Name	Data type	Description
enable	Bool	TRUE: enable functionality of FB
value	Bool	Process value to be forwarded to the redundant channels
valueStatusChannel1	Bool	Value Status of channel 1
valueStatusChannel2	Bool	Value Status of channel 2
twoActuators	Bool	FALSE: configuration single actuator; TRUE: configuration two actuators
acknowledge	Bool	Acknowledge and reintegration after startup or invalid Value Status
ackNec	Bool	TRUE: acknowledge necessary

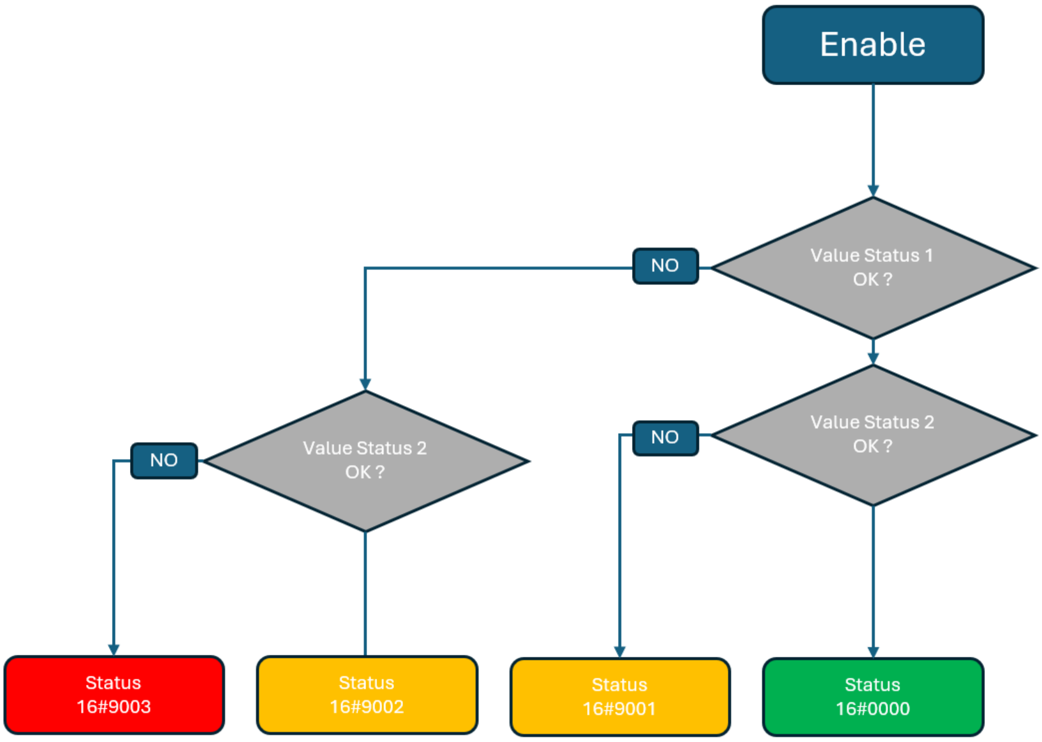
Output parameter

Name	Data type	Description
channel1	Bool	Output value for Channel 1
channel2	Bool	Output value for Channel 2
ackReq	Bool	TRUE: acknowledgement request
valid	Bool	TRUE: valid set of output values available at the FB
busy	Bool	TRUE: FB is running
warning	Bool	TRUE: a channel is invalid
error	Bool	TRUE: an error occurred during the execution of the FB
status	Word	16#0000 - 16#7FFF: Status of the FB, 16#8000 - 16#FFFF: Error identification

Status information

Name	Description
16#0000	Status for both channels are valid
16#7000	No job being currently processed
16#7001	First call after incoming new job (rising edge 'enable')
16#7002	Subsequent call during active processing without further details
16#7003	Block waits for initial acknowledge
16#8600	Error due to an undefined state in state machine
16#9001	Channel 2 passivated
16#9002	Channel 1 passivated
16#9003	Both channels passivated
16#9005	Channel 2 in reintegration process
16#9006	Channel 1 in reintegration process
16#9007	Channel 1 in reintegration process - wait for value = FALSE
16#9008	Channel 2 in reintegration process - wait for value = FALSE

Functional description



Internal block description

State	Comments
Normal operation	When there is no channel error ("**Value Status**" of both channels has the value "TRUE"), then the "**channel1**" and "**channel2**" will be controlled using the value of input "**value**".
Value Status	Even a channel reports an error ("**valueStatusChannel1**" = "FALSE" or/and "**valueStatusChannel2**" = "FALSE"), the affected channel(s) continues to output the process value (input "**value**"), regardless of the Value Status.
Acknowledge ment	As soon as a channel do not report any error anymore, then the output "**ackReq**" is set to "TRUE". A rising edge on the input "**acknowledge**" will acknowledge the error. An acknowledgment is only needed after an error when the parameter "**ackNec**" is set to "TRUE", otherwise no user action is required.

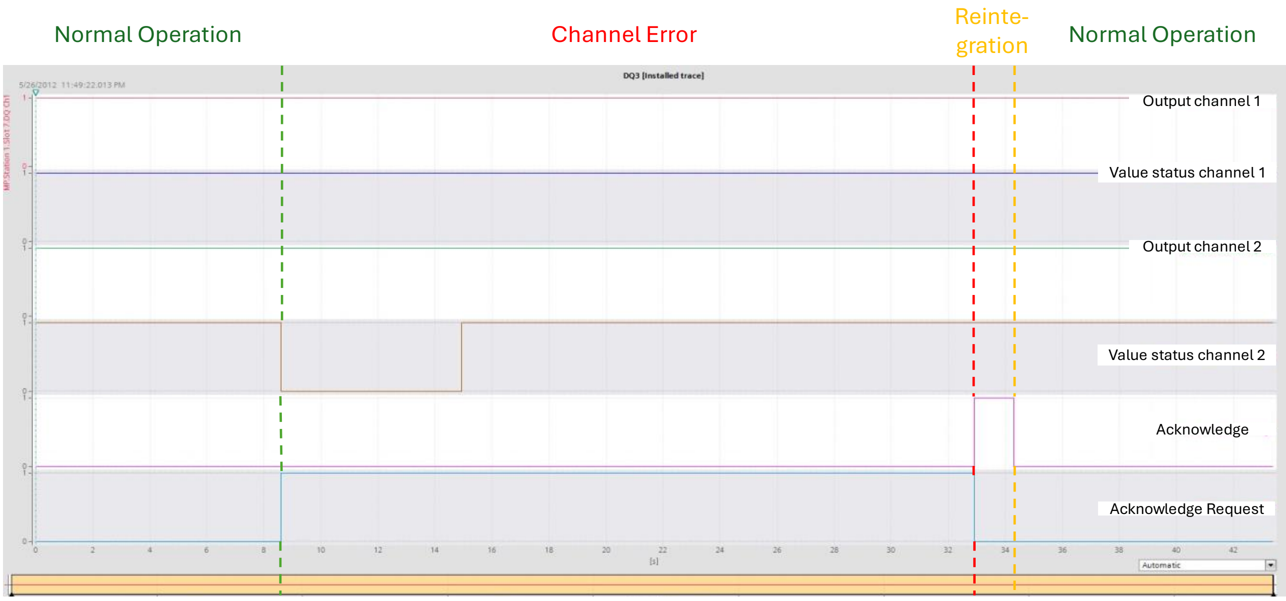
4.5. Reintegration with single actuator

Example demonstrating the reintegration with single actuator after an error.

Configuration with single actuator

Module used	DQ 8x24VDC/0.5A HF	6ES7132-6BF01-0CA0	Firmware V1.0
Diode used	1N4007		
LRedIo_DQ	"**twoActuators**" = FALSE	"**ackNec**" = TRUE	

Phase	Description
Normal Operation	Both channels output the process value, and the Value Status of both channels is 1 (TRUE).
Channel Error	Error on Channel 2, its Value Status switches to 0. The defective channel 2 continues to output the process value regardless of its Value Status, allowing integration without error detection.
Reintegration	When the error of channel 2 is fixed and after reintegration (by an acknowledgement), the function block switches back to normal operation.



5. Redundant Analog Input

5.1. Use cases

Redundant analog input module pair with:

- current-configuration wired to two current-signal sensors,
- current-configuration wired to a single 2-wire-current-signal sensor (additional Zener diodes are required),
- current-configuration wired to a single 4-wire-current-signal sensor (additional Zener diodes are required),
- voltage-configuration wired to two voltage-signal sensors,
- voltage-configuration wired to a single voltage-signal sensor,
- voltage-configuration wired to a single current-signal sensor (through an additional 250 Ohms resistance),

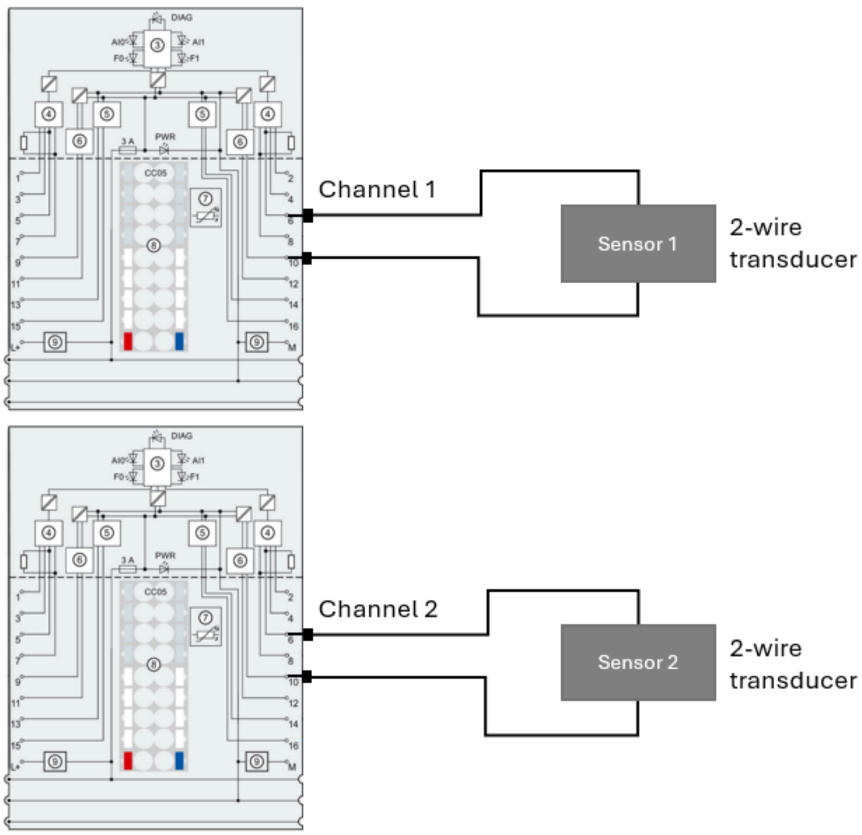
5.2. Wiring configurations

5.2.1. Current-configuration wired to two current-signal sensors

Current measure types supported:

- 0...20mA with 2- or 4-wire transducer
- 4...20mA with 2- or 4-wire transducer
- +/-20mA with 4-wire transducer

Wiring example based on ET 200SP AI 2xU/I 2-/4-wire HF (6ES7134-6HB00-0CA1):



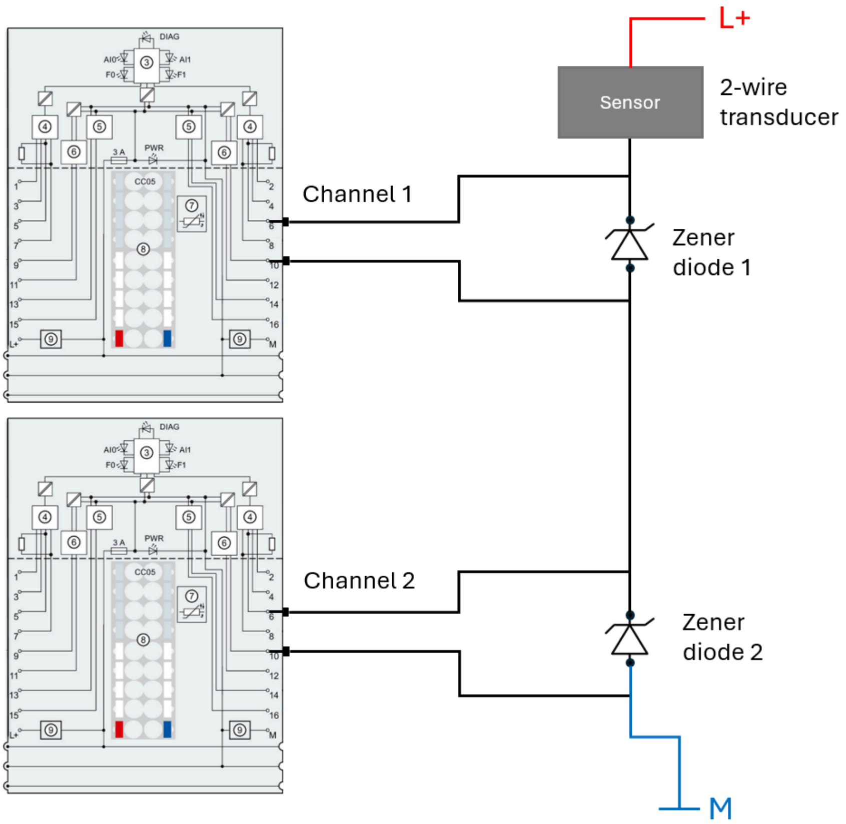
5.2.2. Current-configuration wired to a single 2-wire-current-signal sensor

Current measure types supported:

- 0...20mA with a single 2-wire transducer and additional single Zener diodes
- 4...20mA with a single 2-wire transducer and additional single Zener diodes

Wiring example based on ET 200SP AI 2xU/I 2-/4-wire HF (6ES7134-6HB00-0CA1):

NOTE Since the signal line is powered by an external supply voltage, the channels are configurated in TIA Portal with the measurement type "Current (4-wire transducer)".

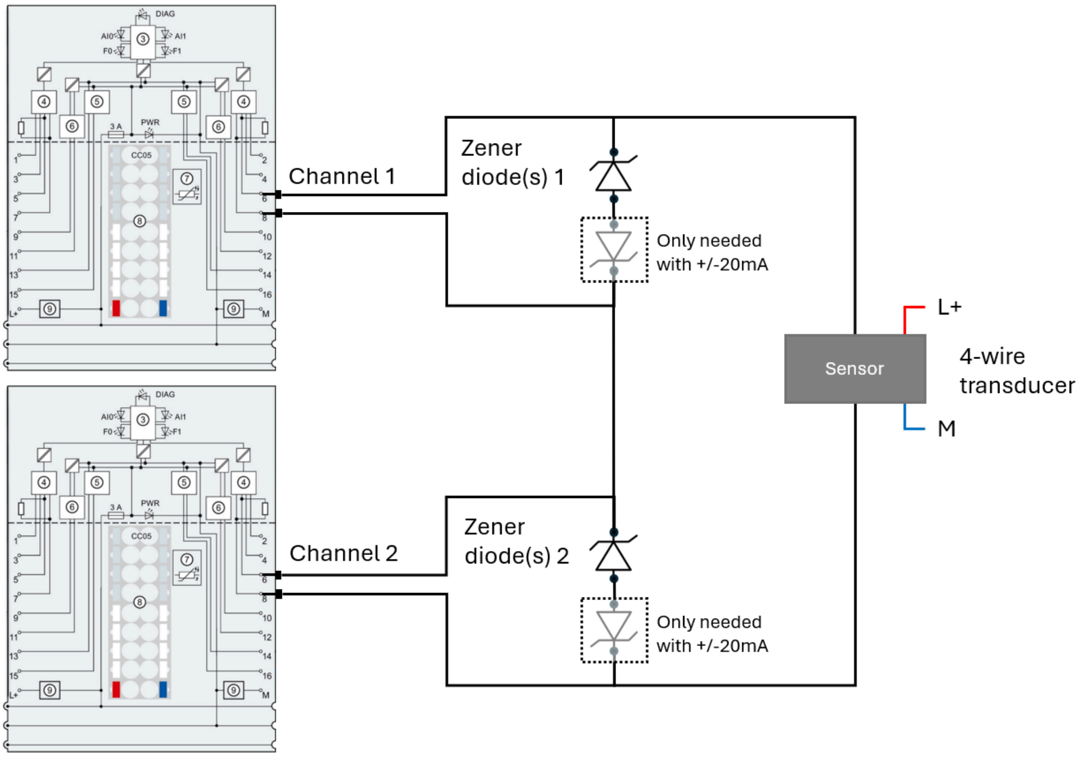


5.2.3. Current-configuration wired to a single 4-wire-current-signal sensor

Current measure types supported:

- 0...20mA with a single 4-wire transducer and additional single Zener diodes
- 4...20mA with a single 4-wire transducer and additional single Zener diodes
- +/-20mA with a single 4-wire transducer and additional anti-serial Zener diodes

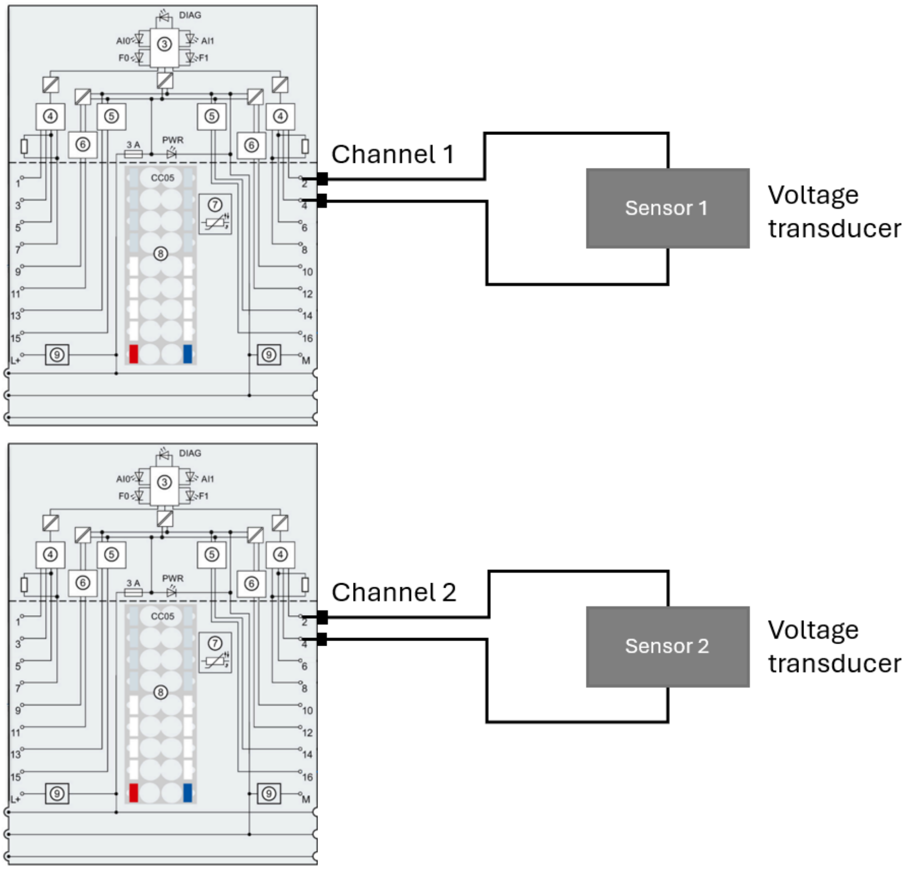
Wiring example based on ET 200SP AI 2xU/I 2-/4-wire HF (6ES7134-6HB00-0CA1):



5.2.4. Voltage-configuration wired to two voltage-signal sensors

Voltage measure types supported: All

Wiring example based on ET 200SP AI 2xU/I 2-/4-wire HF (6ES7134-6HB00-0CA1):

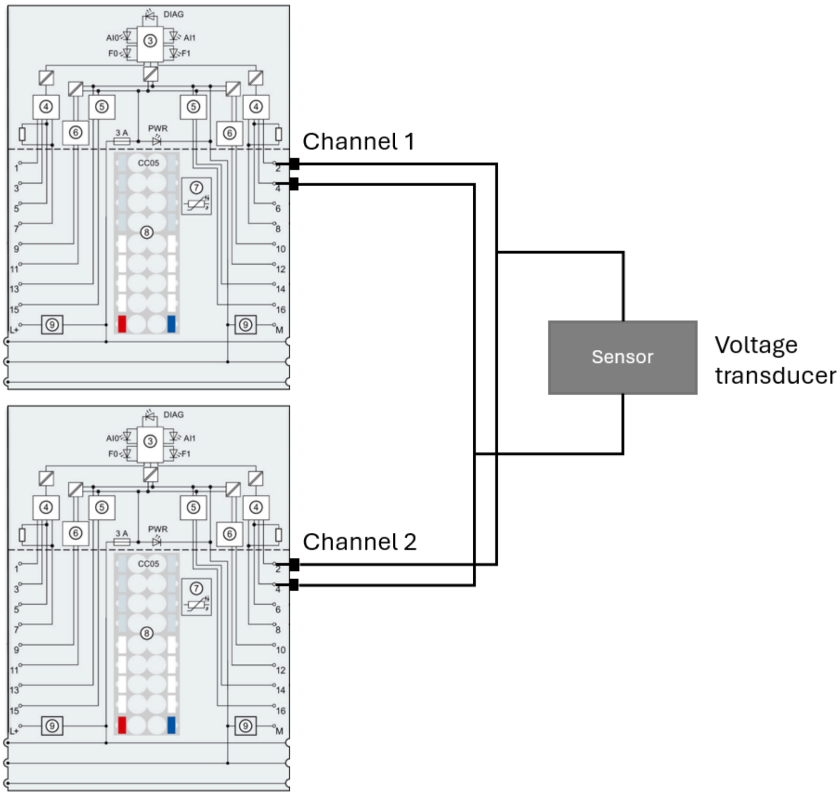


5.2.5. Voltage-configuration wired to a single voltage-signal sensor

Voltage measure types supported:

- +/-10V
- +/-5V
- 0...10V
- 1...5V

Wiring example based on ET 200SP AI 2xU/I 2-/4-wire HF (6ES7134-6HB00-0CA1):



5.2.6. Voltage-configuration wired to a single 4-wire-current-signal sensor

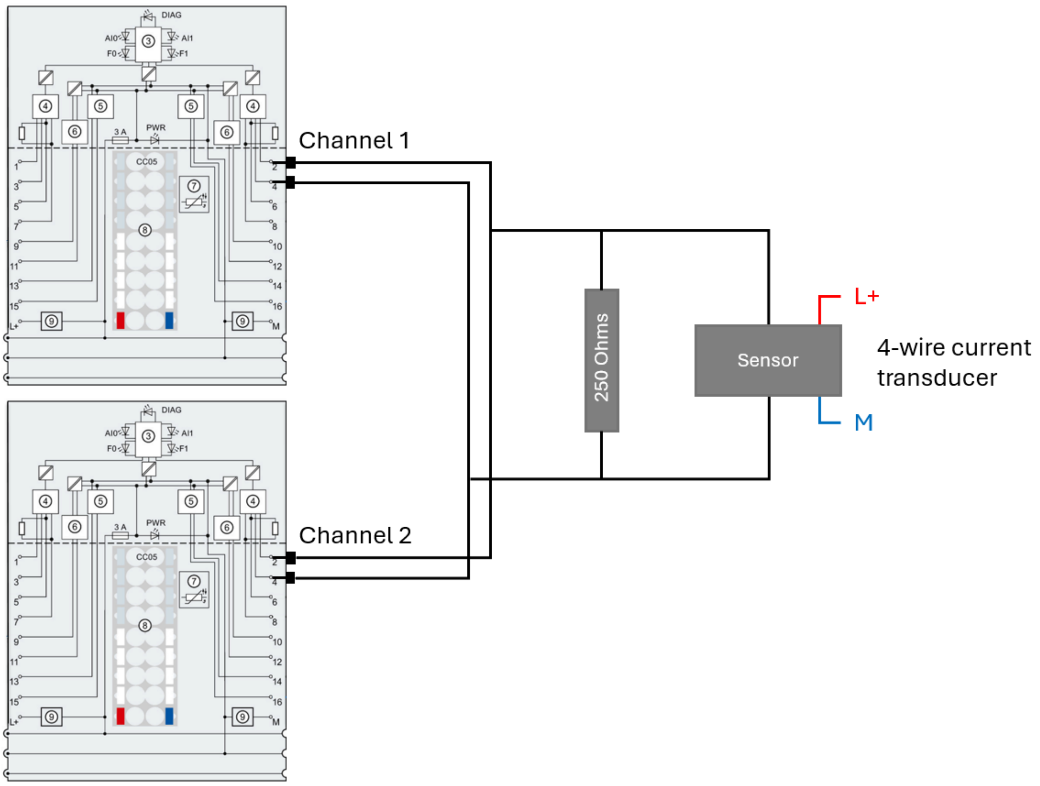
Explanation: an indirect current measurement is performed through an additional 250 Ohms resistance and a voltage configuration of the AI channel (1...5V).

Current	Calculation	Result
4mA	4mA x 250 ohms	1V
20mA	20mA x 250 ohms	5V

Voltage measure type supported:

- 1...5V with 4-wire current transducer

Wiring example based on ET 200SP AI 2xU/I 2-/4-wire HF (6ES7134-6HB00-0CA1):



5.3. Zener diode and current single sensor

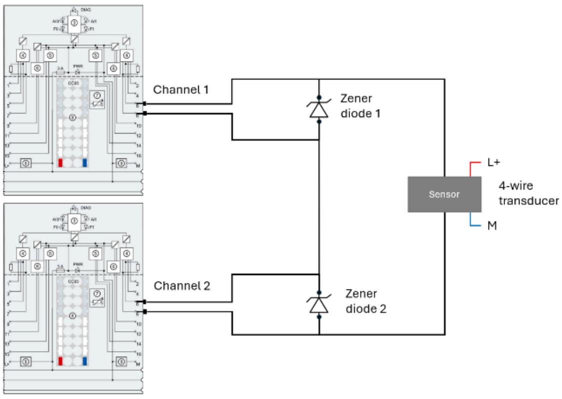
5.3.1. The requirement for Zener diodes

In single sensor current configurations, the components, including AI modules and the sensor, are connected in series. This setup means that if an AI module (either the module itself or a channel) fails, the circuit is interrupted/opened, and the process value becomes unavailable for the controller, thereby affecting the customer’s process.

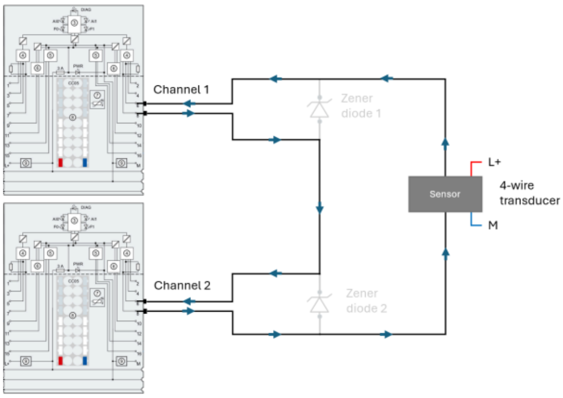
To prevent this and enhance availability, additional Zener diodes are added. Consequently, if a module or channel fails, the current will bypass through the added Zener diode.

See below as example the configuration 4-wire / 4...20mA

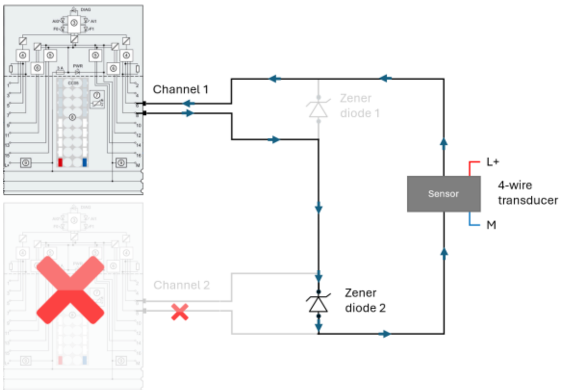
Wiring



Normal operation



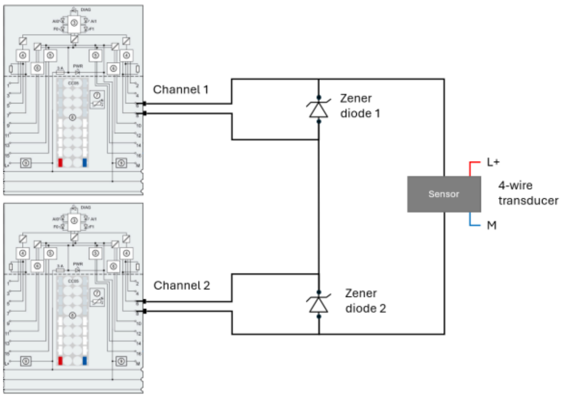
In case of module/channel failure (here as example channel 2)

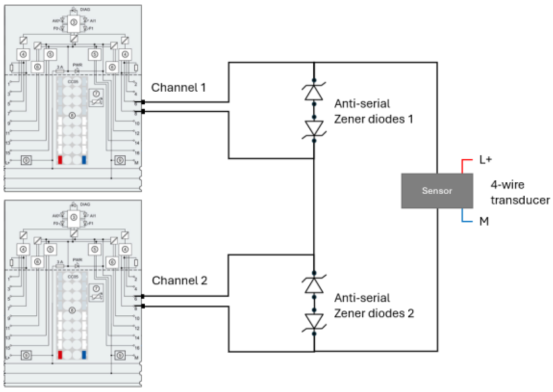


5.3.2. Zener diode setups

Select the right Zener diode setup according to the current measurement and the signal sensor configuration used, either single Zener diode or anti-serial Zener diode setup.

Single Zener diode setup	Anti-serial Zener diode setup
2-wire-current : 0...20mA and 4...20mA	4-wire-current: +/-20mA
4-wire-current : 0...20mA and 4...20mA	





5.3.3. Define the Single Zener diode (0/4...20mA)

To define the right Zener diode, technical data of the AI module are needed (available in Siemens web site): Input resistance, Max output current and when specified the resistance for overvoltage protection by PTC and, diode forward voltage.

Below examples are described in detail:

Step 1: Collect the technical data.

ET 200 Family	MLFB	Description	Input Resistance (Ω)	Diode forward voltage (V)	Max output current (mA)	Resistance for overvoltage protection by PTC (Ω)
ET 200SP	6ES7134-6HB00-0CA1	AI 2 X U/I 2-, 4-Wire HF	130	-	50	-
ET 200SP	6ES7134-6TD00-0CA1	AI 4XI 2-WIRE 4...20MA HART	280	0,35	50	-
ET 200MP	6ES7531-7NF00-0AB0	AI 8xU/I HF	25	-	40	42

Step 2: Calculate the maximum voltage V_{MAX} of the AI channel.

V_{MAX} = (Input Resistance + Resistance for overvoltage protection by PTC) x Max output current

ET 200 Family	MLFB	Description	V_{MAX} (V)
ET 200SP	6ES7134-6HB00-0CA1	AI 2 X U/I 2-, 4-Wire HF	6,5
ET 200SP	6ES7134-6TD00-0CA1	AI 4XI 2-WIRE 4...20MA HART	14
ET 200MP	6ES7531-7NF00-0AB0	AI 8xU/I HF	2,68

Step 3: Select Zener diode: V_{MAX} > Zener Breakdown voltage.

ET 200 Family	MLFB	Description	V_{MAX} (V)		Zener Breakdown voltage (V)	Zener diode type
ET 200SP	6ES7134-6HB00-0CA1	AI 2 X U/I 2-, 4-Wire HF	6,5	>	6,2	ZPD6.2
ET 200SP	6ES7134-6TD00-0CA1	AI 4XI 2-WIRE 4...20MA HART	14	>	8,2	ZPD8B2
ET 200MP	6ES7531-7NF00-0AB0	AI 8xU/I HF	2,68	>	1,8	1N4678

Step 4: Calculate V_{RESULT}.

V_{RESULT} = Zener Breakdown voltage + Diode forward voltage

ET 200 Family	MLFB	Description	Zener Breakdown voltage (V)	Diode forward voltage (V)	V_{RESULT} (V)
ET 200SP	6ES7134-6HB00-0CA1	AI 2 X U/I 2-, 4-Wire HF	6,2	-	6,2
ET 200SP	6ES7134-6TD00-0CA1	AI 4XI 2-WIRE 4...20MA HART	8,2	0,35	8,55
ET 200MP	6ES7531-7NF00-0AB0	AI 8xU/I HF	1,8	-	1,8

Step 5: Check that $I_{RESULT} < I_{MAX}$ and I_{RESULT} > 20mA

$I_{RESULT} = V_{RESULT}$ / (Input Resistance + Resistance for overvoltage protection by PTC)

ET 200 Family	MLFB	Description	V_{RESULT} (V)	I_{RESULT} (mA)		Max output current (mA)
ET 200SP	6ES7134-6HB00-0CA1	AI 2 X U/I 2-, 4-Wire HF	6,2	47,7	<	50
ET 200SP	6ES7134-6TD00-0CA1	AI 4XI 2-WIRE 4...20MA HART	8,55	30,5	<	50
ET 200MP	6ES7531-7NF00-0AB0	AI 8xU/I HF	1,8	26,9	<	40

5.3.4. Define Anti-serial Zener diodes (+/-20mA)

As the previous chapter, to define the right Zener diode, technical data of the AI module are needed (available in Siemens web site): Input resistance, Max output current and when specified the resistance for overvoltage protection by PTC and, diode forward voltage.

Below examples are described in detail:
Step 1: Collect the technical data.

ET 200 Family	MLFB	Description	Input Resistance (Ω)	Diode forward voltage (V)	Max output current (mA)	Resistance for overvoltage protection by PTC (Ω)
ET 200SP	6ES7134-6HB00-0CA1	AI 2 X U/I 2-, 4-Wire HF	130	-	50	-
ET 200SP HA	6DL1134-6AF00-0PH1	AI 8xU/I/TC/4xRTD HA	50	-	40	-
ET 200MP	6ES7531-7NF00-0AB0	AI 8xU/I HF	25	-	40	42

Step 2: Calculate the maximum voltage V_{MAX} of the AI channel.

V_{MAX} = (Input Resistance + Resistance for overvoltage protection by PTC) x Max output current

ET 200 Family	MLFB	Description	V_{MAX} (V)
ET 200SP	6ES7134-6HB00-0CA1	AI 2 X U/I 2-, 4-Wire HF	6,5
ET 200SP HA	6DL1134-6AF00-0PH1	AI 8xU/I/TC/4xRTD HA	2,0
ET 200MP	6ES7531-7NF00-0AB0	AI 8xU/I HF	2,68

Step 3: Select Zener diode: V_{MAX} > Zener Breakdown voltage + 0,7V (anti-serial configuration)

ET 200 Family	MLFB	Description	V_{MAX} (V)		Zener Breakdown voltage (V)		Anti-serial (V)	Zener diode type
ET 200SP	6ES7134-6HB00-0CA1	AI 2 X U/I 2-, 4-Wire HF	6,5	>	5,6	+	0,7	ZPD5.6
ET 200SP HA	6DL1134-6AF00-0PH1	AI 8xU/I/TC/4xRTD HA	2,0	>	1,0	+	0,7	ZPY1
ET 200MP	6ES7531-7NF00-0AB0	AI 8xU/I HF	2,68	>	1,8	+	0,7	1N4678

Step 4: Calculate V_{RESULT}.

V_{RESULT} = Zener Breakdown voltage + Diode forward voltage + 0,7V (anti-serial configuration)

ET 200 Family	MLFB	Description	Zener Breakdown voltage (V)		Diode forward voltage (V)		Anti-serial (V)		V_{RESULT} (V)
ET 200SP	6ES7134-6HB00-0CA1	AI 2 X U/I 2-, 4-Wire HF	5,6	+	-	+	0,7	=	6,3
ET 200SP HA	6DL1134-6AF00-0PH1	AI 8xU/I/TC/4xRTD HA	1,0	+	-	+	0,7	=	1,7
ET 200MP	6ES7531-7NF00-0AB0	AI 8xU/I HF	1,8	+	-	+	0,7	=	2,5

Step 5: Check that $I_{RESULT} < I_{MAX}$ and $I_{RESULT} > 20mA$

$I_{RESULT} = V_{RESULT}$ / (Input Resistance + Resistance for overvoltage protection by PTC)

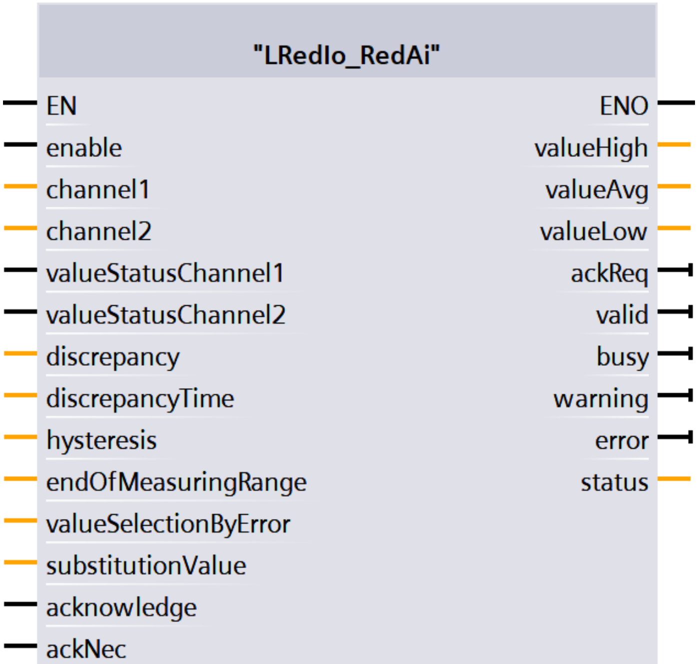
ET 200 Family	MLFB	Description	V_{RESULT} (V)	I_{RESULT} (mA)		Max output current (mA)
ET 200SP	6ES7134-6HB00-0CA1	AI 2 X U/I 2-, 4-Wire HF	6,3	48,5	<	50
ET 200SP HA	6DL1134-6AF00-0PH1	AI 8xU/I/TC/4xRTD HA	1,7	34,0	<	40
ET 200MP	6ES7531-7NF00-0AB0	AI 8xU/I HF	2,5	37,3	<	40

5.4. LRedIo_RedAi

Short description

The function block "LRedIo_RedAi" is used for the evaluation of redundant analog input channels within the user program. According to the process value and the Value Status of each analog input channel, the block provides a process value which is used in the user program. The consistency as well as the quality of both signals are cyclically monitored and therefore deliver valuable diagnostic information of the redundant analog input module pair.

Block interface



Input parameter

Name	Data type	Description
Enable	Bool	TRUE: enable functionality of FB
channel1	Dint	Channel 1
channel2	Dint	Channel 2
valueStatusChannel1	Bool	Value Status of the channel 1
valueStatusChannel2	Bool	Value Status of the channel 2
discrepancy	Real	Discrepancy value between the inputs (min. 0.0% / 0 points - max. 100.0% / 27648 points) Default value: 5.0
discrepancyTime	Time	Max. discrepancy time between the inputs (min. 20ms - max. 100s)
hysteresis	USInt	Hysteresis tolerated after error to allow acknowledgment - specify as a percentage of the discrepancy value (0...100) Default value: 90
endOfMeasuringRange	Dint	End value, of the measuring range normally 27648, overflow value preset, important for discrepancy value
valueSelectionByError	USInt	By error selection of the value used; 0 = Passivated (0); 1 = Substitute value; 2 = Last valid value; 3 = Accept channel values
substitutionValue	Dint	Substitution value
acknowledge	Bool	Acknowledgement and reintegration after startup or invalid Value Status
ackNec	Bool	TRUE: Acknowledgement necessary

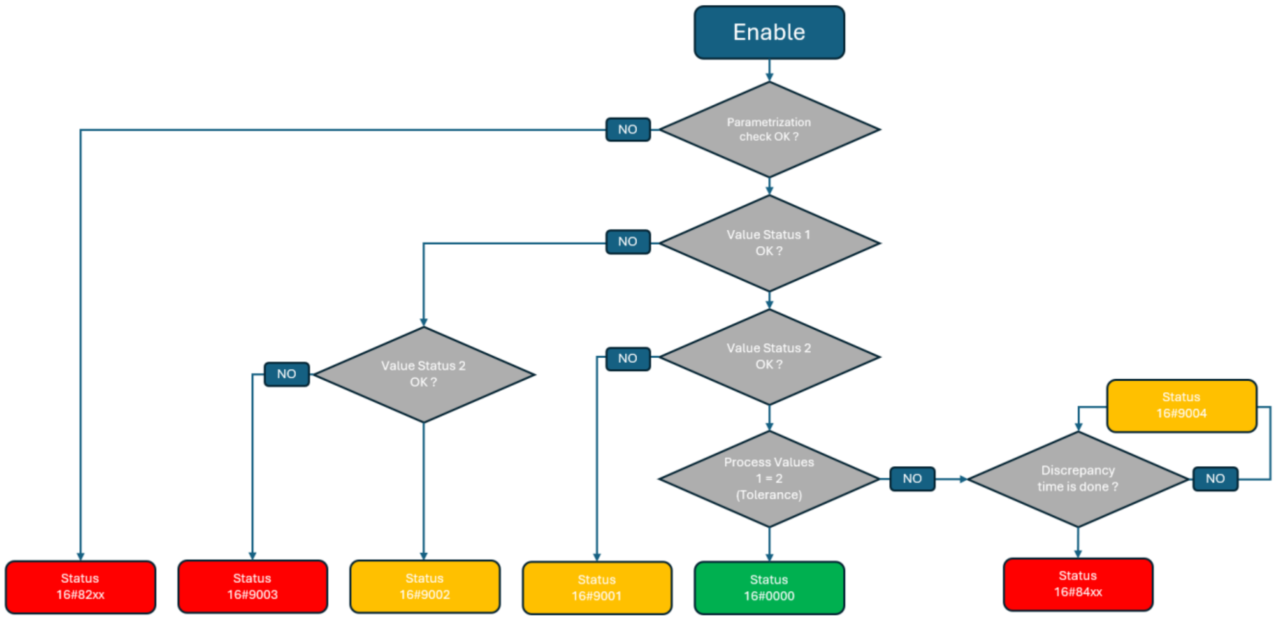
Output parameter

Name	Data type	Description
valueHigh	DInt	Evaluated higher input signal
valueAvg	Dint	Evaluated average input signal
valueLow	DInt	Evaluated lower input signal
ackReq	Bool	TRUE: acknowledgement request
valid	Bool	TRUE: valid set of the value parameter
busy	Bool	TRUE: FB is running
warning	Bool	TRUE: a channel is invalid
error	Bool	TRUE: An error occurred during the execution of the FB
status	Word	16#0000 - 16#7FFF: Status of the FB, 16#8000 - 16#FFFF: Error identification

Status information

Name	Description
16#0000	Both channels are valid
16#7000	No job being currently processed
16#7001	First call after incoming new job (rising edge 'enable')
16#7002	Subsequent call during active processing without further details
16#7003	Block waits for initial acknowledge
16#8201	Parametrization error discrepancy
16#8202	Parametrization error hysteresis
16#8203	Parametrization error end of measuring range
16#8204	Parametrization error substitution value
16#8205	Parametrization error use substitution value
16#8206	Parametrization error discrepancy time
16#8400	Discrepancy error with channel passivation
16#8401	Discrepancy error with substitution value
16#8402	Discrepancy error with last valid values
16#8403	Discrepancy error with using output values (avg, min and max)
16#8600	Error due to an undefined state in state machine
16#9001	Invalid value on Value Status 2
16#9002	Invalid value on Value Status 1
16#9003	Invalid value on both channels
16#9004	Discrepancy between channels but disc time still running

Functional description



Internal block description

State	Comments
Parametrization error	The function block checks if the parametrization is correct. "**discrepancy**" between 0 and 100% "**discrepancyTime**" > 20ms and < 10s "**hysteresis**" between 0 and 100% "**endOfMeasuringRange**" 0 and 32768 "**valueSelectionByError**" between 0 and 3 "**substitutionValue**" between -32512 and 32511 (Min. underrange / Max. overrange value) Otherwise a parametrization error is triggered.
Normal operation	When there is no channel error ("Value Status" of both channels have the value "1") and there is no discrepancy error, then the output values will be controlled using the higher (**valueHigh**), the average (**valueAvg**) and the lower value (**valueLow**) of both channels.
Value Status	When only one of the channels reports an error ("**valueStatusChannel1**" = "0" or "**valueStatusChannel2**" = "0"), then the errorfree "channel" is used for the output values. When both channels report an error ("**valueStatusChannel1**" = "0" and "**valueStatusChannel2**" = "0"), then the output values depend on the parameter "**valueSelectionByError**" defined by the user: "0" = passivated channels (0), "1" = Substitute value, "2" = the Last valid value or "3" = Accept channel values is used.
Behavior by discrepancy	When both channels do not report any error but there is a discrepancy between the "**channel1**" and the "**channel2**" longer than the "**discrepancyTime**", then a warning is triggered. The output values will be set according to the user configuration "**valueSelectionByError**" defined by the user: "0" = passivated channels (0), "1" = Substitute value, "2" = the Last valid value or "3" = Accept channel values.
Acknowledgement	As soon as a channel do not report any error anymore, then the output "**ackReq**" is set to "TRUE". A rising edge on the input "**acknowledge**" will acknowledge the error. An acknowledgment is only needed after an error when the parameter "**ackNec**" is set to "TRUE", otherwise no user action is required.

5.5. Reintegration after discrepancy error

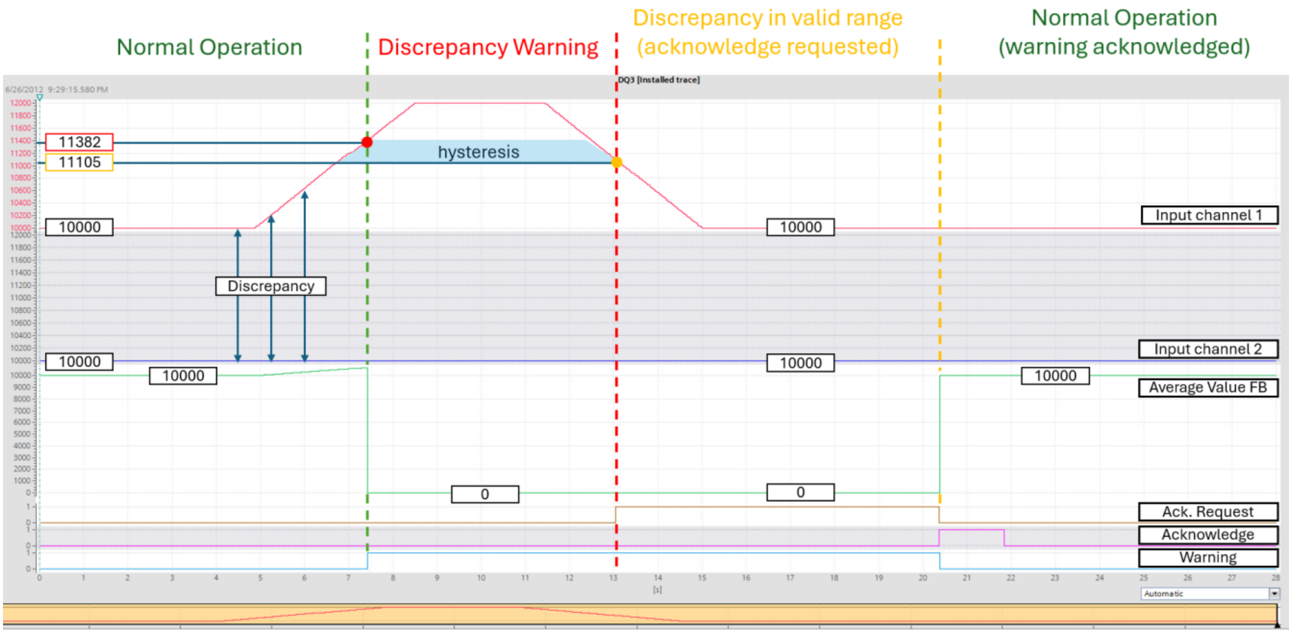
The “**discrepancy**” parameter defines a valid range within the two channel values may vary. If there is a discrepancy outside this defined tolerance, and that, for a time longer than the parameter "**discrepancyTime**", a predefined reaction is triggered by the function block (“0” = passivated channels (FALSE), “1” = Substitute value or “2” = the Last valid value).

After such discrepancy error, and to prevent signal floating, the user can define with the parameter “**hysteresis**” a dedicated acknowledge range. The signal deviation must return into this dedicated range to allow acknowledgment and therefore to reintegrate the process value.

Example: Current-configuration wired to two current-signal sensors

Module used	AI 2xU/I 2-/4-wire HF	6ES7134-6HB00-0CA1	Firmware V2.0	
LRedIo_AI	“**discrepancyTime**” = 50ms “**discrepancy**” = 5.0%	“**hysteresis**” = 80%	“**valueSelectionByError**” = 0	“**ackNec**” = TRUE

Phase	Description
Normal Operation	Both channels are valid, and the values are within the valid range (“**discrepancy** ”, “**discrepancyTime**”).
Discrepancy Warning	[Red circle] The maximum tolerated discrepancy (1382 points) between channels is exceeded (27648 points x parameter “**discrepancy**” 0,05). Therefore, a discrepancy warning is triggered. The “**output value**” (here the Average Value) is passivated (0) accordingly to the parameter “**valueSelectionByError**”.
Discrepancy in valid range (acknowledge requested)	The values get closer to each other and are again within the valid range. The range for acknowledgement is defined by the parameter “**hysteresis**”. As soon as the discrepancy between channels is smaller than 1105 points (80% of 1382 points), an acknowledge request (“**AckReq**”) is triggered [Orange circle] but the output value remains the same (0).
Normal Operation	After the acknowledgement of the warning, the block proceeds with its normal operation.



6. Redundant Analog Output

6.1. Use cases

Redundant analog input module pair with:

- configuration wired to two signal actuators (voltage and current),
- current-configuration wired to a single current-signal actuator (additional diodes are required).

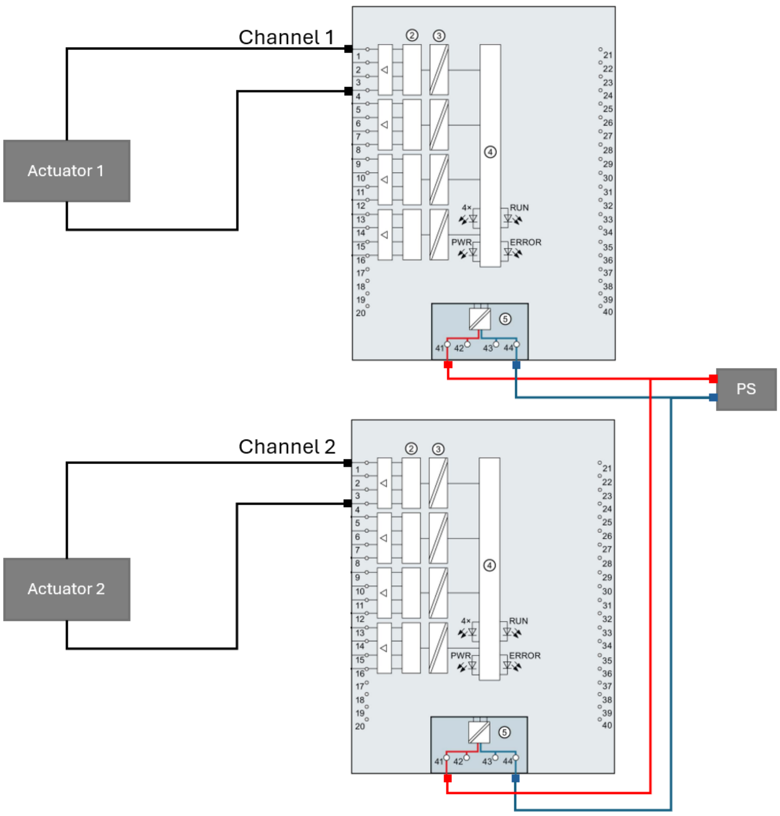
6.2. Wiring configurations

6.2.1. Configuration wired to two actuators

Voltage measure types supported: all

Current measure types supported: all

Wiring example based on ET 200MP AQ 4xU/I HF (6ES7532-5ND00-0AB0):

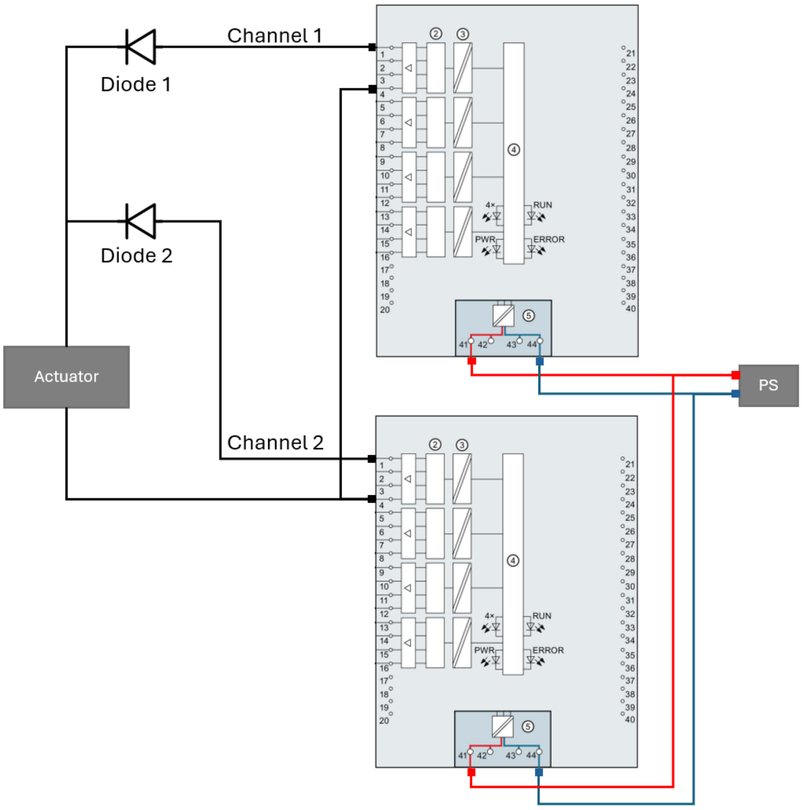


6.2.2. Current-configuration wired to a single current-signal actuator

Current measure types supported:

- 0...20mA with addional standard diodes
- 4...20mA with addional standard diodes

Wiring example based on ET 200MP AQ 4xU/I HF (6ES7532-5ND00-0AB0):



6.3. Single actuator and diode

The table below lists the analog output modules (IP20) which in redundant mode need additional diodes:

ET 200 Family	Additional diode	Comments
ET 200SP	Yes	For example, through terminal blocks
ET 200SP HA	No	included by design into the DQ module
ET 200MP	Yes	For example, through terminal blocks

For ET 200SP or ET 200MP, suitable diodes are with:

- **Voltage Rating:** maximum repetitive peak reverse voltage (V_{RRM}) of > 200V,
- **Current Handling:** forward continuous current (I_F) >= 1A.
- Example: **type 1N4007**

6.4. 4...20mA with single actuator – how to select the right AQ module?

For a configuration 4...20mA with a single actuator, the two following conditions must be met to select the right AQ module:

- An output **current value of 2mA must be supported**.
  Check the underrange of the AQ module in the relevant manual (Table "Current output ranges 4 to 20 mA").

- The **wire break current detection must be <2mA**.
  Check the wire break detection in the relevant manual.
  Even the wire break diagnostic is not enabled, the Value Status reacts as soon as the current reaches the wire break current detection.

Examples:

(From technical data: Current range & Wire break detection / stand 04.2025):

ET 200MP	MLFB	Current range	Wire break detection	Result
AQ 4xU/I HF	6ES7 532-5ND00-0AB0	0 – 22,81mA	0,4mA	OK
AQ 4xU/I ST	6ES7 532-5HD00-0AB0	0 – 22,81mA	0,2mA	OK
AQ 8xU/I HS	6ES7 532-5HF00-0AB0	0 – 22,81mA	3mA	Not suitable

ET 200SP	MLFB	Current range	Wire break detection	Result
AQ4 x I HART	6ES7 135-6TD00-0CA1	0 – 21mA	240µA	OK
AQ2 x U/I HF	6ES7 135-6HB00-0CA1	3,6 – 21mA	3mA	Not suitable
AQ4 x U/I ST	6ES7 135-6HD00-0BA1	3,6 – 21mA	3mA	Not suitable

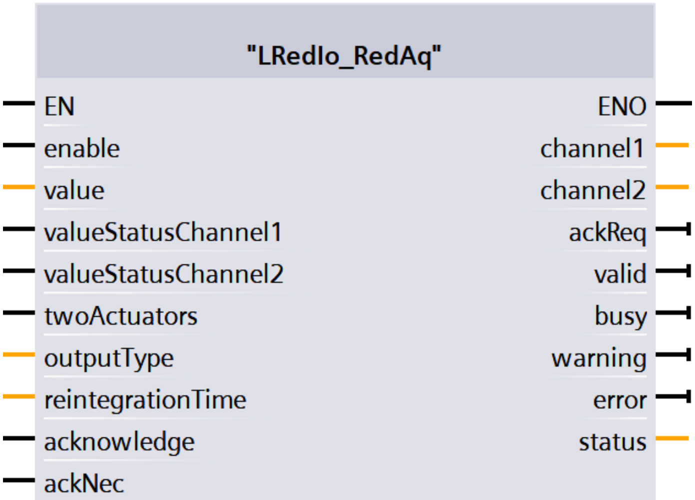
ET 200SP HA	MLFB	Current range	Wire break detection	Result
AQ 8xI HART HA	6DL1135-6TF00-0PH1	0 – 22,81mA	240µA	OK
AQ 4xI HART ISOL HA	6DL1135-6UD00-0PK0	0 – 22,81mA	240µA	OK

6.5. LRedIo_RedAq

Short description

The function block "LRedIo_RedAq" is used for the evaluation of redundant digital output channels within the user program. The block forwards the process value calculated by the user program to both analog output channels. The quality of both signals is cyclically monitored and therefore deliver valuable diagnostic information of the redundant analog output module pair.

Block interface



Input parameter

Name	Data type	Description
enable	Bool	TRUE: enable functionality of FB
value	DInt	Process value to be forwarded to the redundant channels
valueStatusChannel1	Bool	Value Status of the channel 1
valueStatusChannel2	Bool	Value Status of the channel 2
twoActuators	Bool	FALSE: configuration single actuator; TRUE: configuration two actuators
outputType	Int	Signal type of the redundant channel defined in the HW configuration 0: 0...20 mA 1: 4...20 mA 2: ± 20 mA 3: 0...10 V 4: 1...5 V 5: ± 5 V / ± 10 V
reintegrationTime	Time	Time for the active and passivated channels to 50% - only necessary for single actuator configuration
acknowledge	Bool	Acknowledgement and reintegration after startup or invalid Value Status
ackNec	Bool	TRUE: Acknowledgement necessary

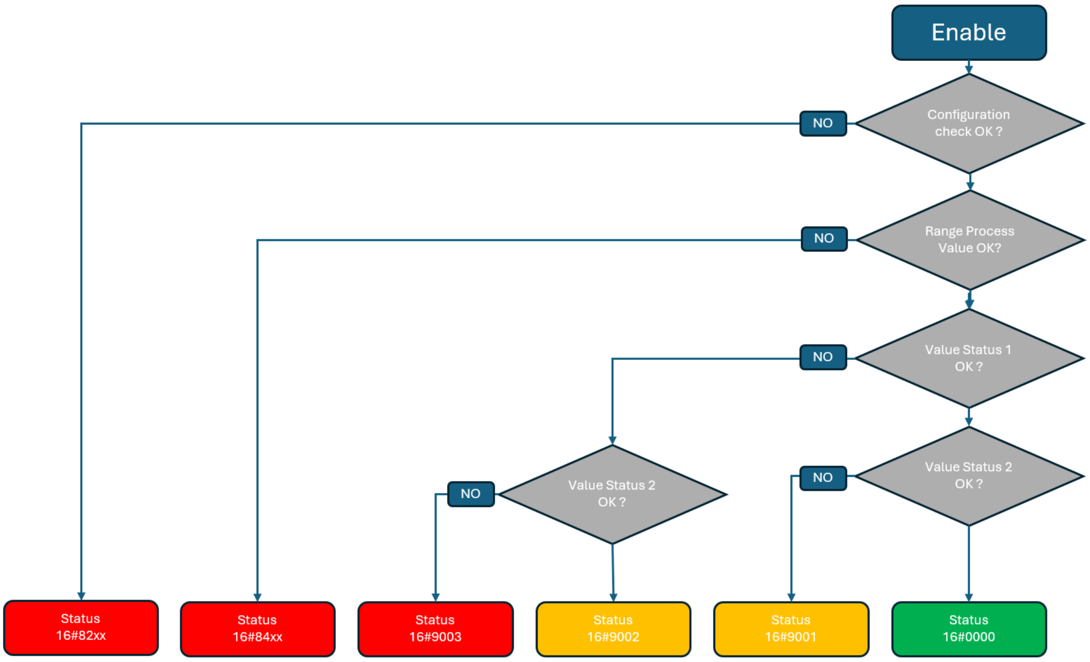
Output parameter

Name	Data type	Description
channel1	Int	Output value channel 1
channel2	Int	Output value channel 2
ackReq	Bool	TRUE: acknowledgement request
valid	Bool	TRUE: output signals are valid
busy	Bool	TRUE: FB is running
warning	Bool	TRUE: a channel is invalid
error	Bool	TRUE: An error occurred during the execution of the FB
status	Word	16#0000 - 16#7FFF: Status of the FB, 16#8000 - 16#FFFF: Error identification

Status information

Name	Description
16#0000	Both channels are valid
16#7000	No job being currently processed
16#7001	First call after incoming new job (rising edge 'enable')
16#7002	Subsequent call during active processing without further details
16#7003	Block waits for initial acknowledge
16#8200	Enum for signal type out of range
16#8207	Reintegration time is too low
16#8208	Single actuator with signal type voltage not allowed
16#8407	Process value in overflow area
16#8408	Process value in underflow area
16#8600	Error due to an undefined state in state machine
16#9001	Channel 2 passivated
16#9002	Channel 1 passivated
16#9003	Both channels passivated
16#9005	Reintegration phase - channel 1 active, channel 2 passive
16#9006	Reintegration phase - channel 2 active, channel 1 passive

Functional description



Internal block description

State	Comments
Parametrization error:	The function block checks if the parametrization is correct. "**outputValue**" between 0 and 5 **"reintegrationTime"** > 50ms - only necessary for single actuator configuration Otherwise a parametrization error is triggered.
Normal operation	There is no channel error reported ("Value Status" of both channels have the value "TRUE"). The output value of channel 1 and 2 depends on the parameter "**twoActuators**". If there are two actuators ("**twoActuators**" = TRUE), then each output "**Channel1**" and "**Channel2**" will delivers 100% of the required process "**Value**". If there is a single current actuator ("**twoActuators**" = FALSE), and because the currents are added, then each output "**Channel1**" and "**Channel2**" will delivers 50% of the required process "**Value**".
Value Status	When only one of the channels reports an error ("**valueStatusChannel1**" = "FALSE" or "**valueStatusChannel2**" = "FALSE"), then the errorfree "channel" delivers 100% of the required process "**Value**". When both channels report an error ("**valueStatusChannel1**" = "FALSE" and "**valueStatusChannel2**" = "FALSE"), then the output value is set to 0.
Acknowledgement	As soon as a channel do not report any error anymore, then the output "**ackReq**" is set to "TRUE". A rising edge on the input "**acknowledge**" will acknowledge the error. An acknowledgment is only needed after an error when the parameter "**ackNec**" is set to "TRUE", otherwise no user action is required.

6.6. Reintegration after error with single current actuator

For the configuration with a single current actuator, if one channel fails, then the errorfree one will take over 100% of the required current. When the failure is fixed and after reintegration (by an acknowledgement), a dedicated adjustment process is carried out to prevent more than 100% current being applied to the actuator.

This adjustment allows the analog outputs to ramp up and down gradually so that each analog output provides again 50% of the required current and therefore the required process "**Value**".

The duration of this adjustment is defined by the parameter "**reintegrationTime**".

Example with the scenario "Channel 1 fails":

	Channel 1	**Channel 2**	**Result**
Normal operation	50%	50%	100%
Channel 1 fails	0%	100%	100%
Channel 1 fixed and Reintegrated	0%	100%	100% (**Start** of the reintegration)
Way to reach normal operation	Ramp up 0% ➔ 50%	Ramp down 100% ➔50%	100% (Reintegration in progress)
-	0%	100%	100% (Reintegration in progress)
-	1%	99%	100% (Reintegration in progress)
	2%	98%	100% (Reintegration in progress)
-	…	…	100% (Reintegration in progress)
-	48%	52%	100% (Reintegration in progress)
-	49%	51%	100% (Reintegration in progress)
-	50%	50%	100% (**End** of the reintegration)
Normal operation	50%	50%	100%

Example: Current-configuration wired to a single current-signal actuator

Module used	AQ 4xU/I HF	6ES7532-5ND00-0AB0	Firmware V1.1
Diode used	1N4007		
LRedIo_AQ	"**twoActuators**" = FALSE	"**outputType**" = 1	"**ackNec**" = TRUE

Phase	Description
Normal Operation	Both channels deliver 50% of the process value. The Value Status of both channels is 1 (TRUE).
Channel Error	Error on "**Channel 2**", its Value Status switches to 0. The "**Channel 1**" takes over the 100% of the process value.
Reintegration	When the failure is fixed and after reintegration (by an acknowledgement), a dedicated adjustment process is carried out. It allows Channel 2 to ramp up, and Channel 1 to ramp down gradually so that each analog output provides again 50% of the required current. The duration of this adjustment is defined by the parameter "**reintegrationTime**".

7. Appendix

7.1. Service and support

SiePortal

The integrated platform for product selection, purchasing and support - and connection of Industry Mall and Online support. The SiePortal home page replaces the previous home pages of the Industry Mall and the Online Support Portal (SIOS) and combines them.

- Products & Services
  In Products & Services, you can find all our offerings as previously available in Mall Catalog.
- Support
  In Support, you can find all information helpful for resolving technical issues with our products.
- mySieportal
  mySiePortal collects all your personal data and processes, from your account to current orders, service requests and more. You can only see the full range of functions here after you have logged in.

You can access SiePortal via this address: sieportal.siemens.com

Technical Support

The Technical Support of Siemens Industry provides you fast and competent support regarding all technical queries with numerous tailor-made offers – ranging from basic support to individual support contracts.
Please send queries to Technical Support via Web form: support.industry.siemens.com/cs/my/src

SITRAIN – Digital Industry Academy

We support you with our globally available training courses for industry with practical experience, innovative learning methods and a concept that's tailored to the customer's specific needs.
For more information on our offered trainings and courses, as well as their locations and dates, refer to our web page: siemens.com/sitrain

Industry Online Support app

You will receive optimum support wherever you are with the "Industry Online Support" app. The app is available for iOS and Android:

7.2. Links and literature

No.	Topic
\1\	Siemens Industry Online Support https://support.industry.siemens.com
\2\	Link to this entry page of this application example https://support.industry.siemens.com/cs/ww/en/view/109767576
\3\	What are the differences between the Basic (BA), Standard (ST), High Feature (HF) and High Speed (HS) modules of the ET 200SP and ET 200MP? https://support.industry.siemens.com/cs/ww/en/view/109476914

7.3. Change documentation

Version	Date	Modification
1.0	06/2019	First version
2.0	05/2025	Library "LRedIo" updated (all Function Blocks) Configurations single analog sensor to Analog Input module pair added

